

# Swiss Re SONAR

New emerging risk insights



Every year, Swiss Re SONAR informs and inspires conversations about emerging risks, helping the insurance industry and its clients continue to build resilience.

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# Emerging risk themes

Potential impact

High

0 – 3 years

> 3 years

Medium

Low



Exclusive markets – when geopolitics polarises business



AI getting hacked – systemic vulnerabilities of a booming technology



Bird flu – wake-up call for outbreaks of animal diseases?



Solar radiation management – risks from reversing climate change



The global debt distress – fallout from rising interest rates



Longer lives – impact of medical advances on life expectancy



Accidental or hostile – dangers from nuclear radiation release



Digital payments – disruption risks in a cashless economy



Toxic chemicals – contamination claims on the horizon



Sustainable recycling – emerging liabilities in the circular economy



Climate change, trade and geopolitics – new risks in the Arctic



Weight-loss injections: miracle drugs or new risks in a bottle?



Hooked online – anxiety and distraction from social media binging

Most affected business areas



Property lines



Specialty lines



Casualty



Life & Health



Financial markets  
incl. insurers' assets



Operations  
incl. regulatory changes

## Trend spotlights



- Overlooked exposures: emerging risk drivers for Life & Health insurers
- Digital twins: virtual product testing and maintenance
- Generative AI
- Going digital: embedded insurance



## Foreword

We live in turbulent times. Geopolitical polarisation continues, wars are ongoing and threats of nuclear events have become daily news. Such concerns remind us of a world we thought long gone with the demise of the Cold War. At the same time, recent bank sector woes and ripple effects through the financial markets bring back memories of the global financial crisis of 2008–09.

New and recurring risks foster accumulation of uncertainty, and come alongside long-term trend threats such as climate change. They coincide in a post-pandemic environment, where economic growth has been shaken. Turbulent times are those where expectations are challenged, and predictions fail. These times tax the resilience of individuals, organisations and society at large.

Emerging risks are new or changed risks that are unfamiliar and subject to debate. This is the case, for instance, with recent technological innovations such as generative artificial intelligence. It also applies to changes in legal and regulatory environments, and how society will adapt to rapid digitalisation.

SONAR emphasises scenario thinking to better grasp the uncertainties associated with specific emerging risks. Historical experience, events and contextual circumstances cannot be applied directly to new phenomena. However, consideration of what happened in the past can support and inform mitigation thinking with respect to new emerging risks.

This year's SONAR report features new emerging risk insights and shares findings from Swiss Re's foresight process with the insurance industry and the public at large. To proactively engage our clients and industry stakeholders in discussion of emerging risks is the best way to prepare, mitigate and transform them into new risk pool opportunities.



A handwritten signature in black ink that reads "Patrick Raaflaub". The signature is fluid and cursive, with the first letters of the first and last names being capitalized and prominent.

Patrick Raaflaub  
Group Chief Risk Officer

# Introduction

The world undergoes constant change, and this gives rise to what we term “emerging risks”. These are new and changing risks that are difficult to quantify, yet can have significant impact on the insurance industry. Swiss Re identifies emerging risks by gathering signals and feedback from underwriters, client managers, risk experts and others across the Group, and also from external specialists and research institutions. With the annual SONAR report, published since 2013, we aim to foster dialogue with all stakeholders to better understand and manage emerging risks.

This year’s SONAR report features 13 new emerging risk themes and four emerging trend spotlights. The emerging risk themes are potential new or changing risks, with both downside risks and upside potential for the insurance industry. The “trend spotlight” items highlight contextual developments we deem relevant for insurers, without necessarily profiling a specific risk.

The emerging risk themes outlined in the SONAR report are based on early signals collected over the course of a year. They do not reflect entire industry-wide thinking with respect to emerging risks, nor do they necessarily cover the full list of associated topics currently on Swiss Re’s radar screen. The themes have been categorised according to their estimated impact and our projections for when they may materialise. They are further grouped according to lines of business where we think the biggest exposures will lie (see page 4).

For all the risk themes, we flag where there have been topic-relevant mentions in previous SONAR reports. Some emerging risks presented in this year’s SONAR report may never materialise, while others may form the basis for future risk pools. Similarly, some of the trends depicted may lose importance, while others may play an evolving and growing role in shaping the future business environment.

We begin the SONAR report with an overview of the macro trends relevant to re/insurance markets and the world at large, as seen by Swiss Re. These trends and their grouping into demographic & social, political & economic, technological & natural, and competitive & business themes serve as a backdrop and shape the structure of this year’s report on emerging risk themes and trend spotlights. The appendix explains all terms & definitions used in the report.

Per lines of business, the top emerging risk themes identified in this year’s edition are:

- for Property:  
“Solar radiation management – risks from reversing climate change”
- for Speciality:  
“Exclusive markets – when geopolitics polarises business”
- for Casualty:  
“Toxic chemicals – contamination claims on the horizon”
- for Life & Health:  
“Longer lives – impact of medical advances on life expectancy”
- for Financial markets (including insurers’ assets):  
“The global debt distress – fallout from rising interest rates”
- for Operations (incl. regulatory changes):  
“AI getting hacked – systemic vulnerabilities of a booming technology”

# Macro trends

Identifying and monitoring macro trends enhances our understanding of the risk landscape of the future. Based on internal surveys and workshops, Swiss Re curates a set of macro trends deemed to be of high importance for the insurance industry in the next decade. The macro trends portfolio informs strategic priorities.

This year's updates include three new macro trends: one on the global demographic shift of an ageing world population ("Shifting demographics"; see also the overarching theme "Demographic challenges" below); another on the

continued prevalence of mental health issues, these compounded by lockdowns as a result of the pandemic and post-pandemic conditions (see also overarching theme "A new post-pandemic world" below); and one reflecting on the new disruptive heights of digitalisation in "Impact of generative AI" (see also the theme below on "The rise of the digital economy").

Below is an overview of all macro trends, followed by presentation of the three overarching themes connecting this year's SONAR topics.

## Demographic and social environment

- Shifting demographics
- Growing middle class in high growth markets
- Longevity & radical medical innovation
- Prevalence of mental health issues
- Mass migration & urbanisation
- Changing workplace and talent gaps
- Rising social inequality & unrest

## Political and economic environment

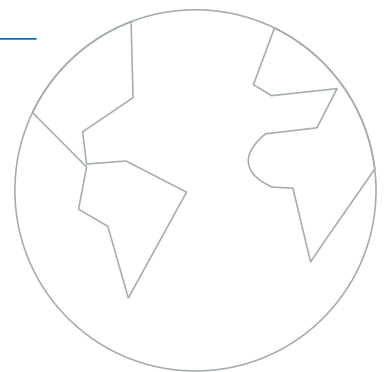
- Macroeconomic fragility
- Challenged globalisation
- Geopolitical & economic instability
- Rising interest rates and risk of persistent inflation
- Infrastructure funding needs

## Technological and natural environment

- Addressing physical climate change risks
- Rising importance of biodiversity and ecosystem services
- Transition to a low carbon economy
- Expansion of digital & cyber risk
- Data as an asset
- Impact of generative AI
- Digital products and processes
- Disruptive digital technologies
- Autonomous transportation & robotics

## Competitive and business environment

- Re/insurance value chain disaggregation and rise of alternative re/insurance providers
- Consolidation of platforms as a business model through strategic partnerships
- Regional champions going global
- Increasing digital customer interaction
- Increasingly litigious environment
- Rising importance of Environmental, Social and Governance (ESG)



## Demographic challenges

The world is undergoing fundamental demographic shifts, most notably in the form of a rapidly ageing population. By 2035, it is expected that there will be more than 1.1 billion people aged over 65 globally, up from only 727 million in 2020 – an expected increase of almost 60% in just 15 years.<sup>1</sup> However, not all regions and countries will age at equal pace. For instance in China, the population is ageing at an unprecedented speed. In regions with already large portions of older people such as Europe and Japan, the overall population will also continue to age, but at a more modest pace. On the other hand, it is expected that most countries in Africa will not see notable population ageing before 2035. Globally, other trends expected to continue are migration and urbanisation as high-growth markets see ongoing expansion of their middle classes.

For insurers, expanding life spans bring both new questions and new risk pool opportunities. One of the main challenges that insurers will face is the uncertainty around the outcome of innovations in drug development and care technology, making it difficult to accurately predict mortality and morbidity trends (see “Longer-lives – impact of medical advances on life expectancy,” p 12). While life expectancy may be growing, *healthy* life expectancy is not increasing at the same pace. Older age cohorts can expect an increase in morbidities, and modelling this dynamic in parallel with ongoing medical advances will be a key challenge for L&H insurers.

While some trends such as medical innovation can be considered positive risk drivers for life expectancy, actuarial modelling will also need to account for emerging negative drivers for life expectancy and quality of life. Over the last 10 years of SONAR research, Swiss Re has identified more than 50 distinct signals and trends with potential to impact L&H insurance, positively and negatively (see “Overlooked exposures: emerging risk drivers for L&H insurers – Trend spotlight,” p 14). Many emerging risks in L&H are not related to medical developments but could be driven by other factors like climate change or environmental pollution (see “Toxic chemicals – contamination claims on the horizon,” p 38). In this regard, L&H insurers will need to look beyond the traditional focus of health-related data to address additional new risk drivers. They will also need to take into account wider societal implications of demographic changes like shifts in medical care spending, saving patterns and the structure of working populations. These will have significant effect on care and healthcare spending in later life, creating many challenges for insurers and societies alike.

## A new post-pandemic world

The COVID-19 pandemic may be over but its scars remain very visible. On the economic front, rising interest rates that followed in the wake of post-pandemic inflationary pressures bring into question the sustainability of sovereign debt across the world (see “The global debt distress – fallout from rising interest rates”, p 20). Further, supply chain disruptions during the pandemic, coupled with rising geopolitical tensions, have altered trading routes and offshoring patterns. These are contributing to the seeming development of a new world order, one of greater polarisation and mutually exclusive markets, which would impede global businesses operations, trade and capital flows (see “Exclusive markets – when geopolitics polarizes businesses”, p 22). Risk of military conflict is high, with potentially catastrophic consequences, including possibly even a nuclear event (see “Accidental or hostile – dangers from nuclear radiation release”, p 26).

On the health front, many people across the world are still grappling with the effects of long COVID, and uncertainties regarding the long-term health implications thereof remain. Several countries continue to report excess mortality, still higher than in pre-pandemic times.<sup>2</sup> In parallel, for many the mental health issues triggered by lockdowns, particularly among young people, linger.<sup>3</sup>

All the while, there is concern about the potential for new pandemics. The recent outbreaks of Avian influenza, the largest to date, in different parts of the world (see “Bird flu – wake-up call for outbreaks of animal diseases?”, p 16) only fuel ongoing worries about global health conditions. Further, questions remain over the lessons learned from the COVID-19 experience and whether societies and insurers are prepared for a next pandemic, whenever that may come, and in whatever shape it may take.

<sup>1</sup> “Living Longer Around the World: Opportunities and Challenges”, WDA Forum, April 2022.

<sup>2</sup> “An enigmatic illness: Long Covid and its implications”, Swiss Re, March 2023.

<sup>3</sup> See “Out of sight, out of mind – mental health issues among the young,” Swiss Re SONAR 2020, p 20.



## The rise of the digital economy

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Digitalisation of the economy is advancing at unprecedented speed. Technological breakthroughs in fields ranging from artificial intelligence (AI) to smart mobility have the potential to fundamentally alter society and way of living, with implications for labour markets and legal systems, among others. For insurers, the spread of digital economy brings many benefits. For example, the integration of digital twins across a wide range of industries is yielding cost-savings in product testing, design and maintenance, and a possible promise of reduced product liability claims in the future (see “Digital twins: virtual product testing and maintenance”, p 46). The switch to digital platforms also offers opportunities for new insurance avenues, such as embedded insurance (see “Going digital – embedded insurance,” p 54). With digitalisation, partnerships between insurers, tech companies and non-insurers have led to insurance offerings that can be seamlessly added to online products or services. Insurers may also benefit from such bundled, digital offerings by deriving improved risk insights from usage-based and behavioural data. Risk pricing improvements can be turned into more affordable offerings.

However, as new digital technologies shape an ever-growing number of industries, the scope for new vulnerabilities and disruption risks will also grow, ultimately affecting the claims experience of insurers. Further, the increasing use of Machine Learning (ML) systems will expose a growing number of services to potential hacking attacks in pursuit of sabotage, fraud, theft and for other purposes (see “AI getting hacked – systemic vulnerabilities of a booming technology”, p 32). Preventing adversarial attacks against ML systems will become a main concern for businesses, insurers and regulators alike. Some cyberattacks could have the potential to paralyse large sectors of the economy and bring with it a rise in liability claims, for instance if the attacks result in failure of digital payment systems (see “Digital payments – disruption risks in a cashless economy”, p 44).

Digitalisation also raises important legal questions with respect to data ownership and liability. The rise of generative AI technology that can create a vast range of content from sources available on the internet, brings concerns with regards to copyright, intellectual property and privacy infringement (see “Generative AI – Trend spotlight”, p 48). Further, for firms operating in the digital services field, the development of new legal frameworks such as the Digital Services Act (DSA) in the European Union (EU), could lead to a new liability risk landscape. There are also increasing concerns over the mental health impacts of online platforms, particularly when usage thereof becomes addictive. More evidence in this field may bring increasing liabilities for social media platform providers (see “Hooked online – anxiety and distraction from social media binging”, p 30).









# Demographic and social environment

# Longer lives – impact of medical advances on life expectancy

Several advances in medicine are leading to longer lives, changing mortality expectations and creating new insurance opportunities.

## Potential impacts

- Uncertainty around the outcome of innovations in disease detection and diagnostics, drug development and care technology among others make it more difficult to accurately predict mortality and morbidity trends.
- Modelling for Life & Health (L&H) lines will be impacted, as will sector profitability forecasts.
- Medical malpractice and product liability claims for new therapies/treatments designed to extend lifespan are possible.

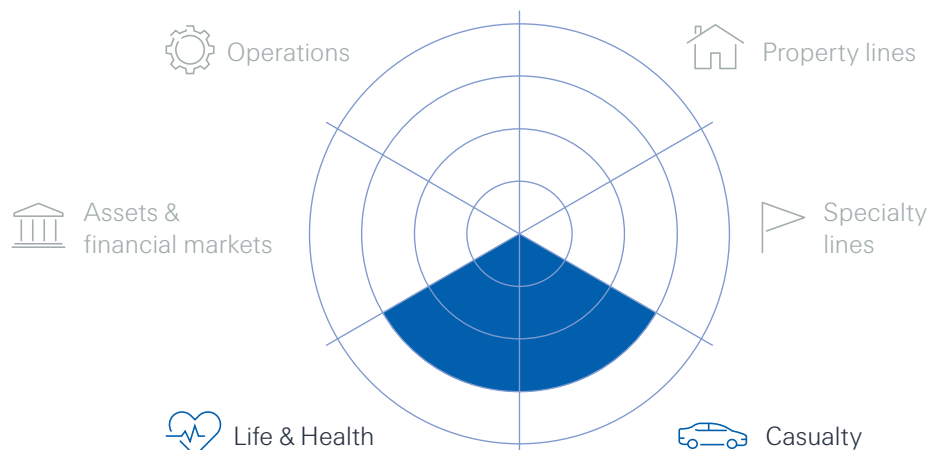
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



Global average life expectancy has increased by almost 23 years over the past 60 years. Between 1955–60 to 2015–20, life expectancy at birth increased from 49.4 to 72.3 years, an average annual gain of 0.5% (or 3.7 months) per person per year. If the demographic trends of the last 20 years persist, the 21st century will be one of population ageing. The number of older persons, in absolute terms and as a share of the world's population, will continue to rise. The expectation is that by 2050, one in six people in the world will be over the age of 65, compared with 1 in 11 in 2019. The number of over-65s globally is forecast to rise from 0.7 billion in 2019 to 1.5 billion in 2050, an increase of 120%.<sup>4</sup>

## Extending lifespans

Longer life expectancy results from improvements in infant mortality rates and overall health conditions for society at large, including better understanding of

underlying risk factors like genetics, lifestyle impacts, modifications to lifestyle behaviours and improvements in access to healthcare.

Medical progress continues, with new therapies for illnesses such as Alzheimer's<sup>5</sup> and osteoporosis<sup>6</sup>, which impact older generations most. In addition, developments triggered by interest in body enhancements, anti-ageing and life extension treatments advocated by transhumanism are gaining more attention.<sup>7</sup> However, as people age, they are also more vulnerable to comorbidities that can span a broad range, including musculoskeletal ailments (eg, osteoarthritis) dementia, pulmonary disease and diabetes.

## Financing longer lives

Age-related and chronic illnesses comprise a large part of today's healthcare provisions in developed markets. Episodic hospital and speciality care at the end of life often

dominate as cost drivers in developed health systems. This has triggered increased need for public funding/support of healthcare in many markets. Rising costs of healthcare can also translate into higher premiums for associated insurance covers.<sup>8</sup>

For life insurance, modelling for old age will become more challenging as medical innovations alter life expectancies. This will have knock-on effects on longevity solutions. For instance, L&H insurance services may also include options to finance care for the elderly in their own homes rather than in hospitals or other institutional care settings.<sup>9</sup>

The development of new treatments for poor health in old age will continue, too. Given their typically more fragile constitutions, the elderly can be more vulnerable than other age groups to sustain adverse reaction to new medicines or treatments. This raises the potential for product liability and medical malpractice claims, to which casualty insurers would be most exposed.

#### Related articles in SONAR

- “Biotech boom drives innovation risk”, SONAR 2022
- “Baby dreams – risk in the fertility services”, SONAR 2022
- “COVID-19 – the longer-term health burden”, SONAR 2021
- “What’s next in insurance modelling?”, SONAR 2021
- “The fragility of public healthcare”, SONAR 2020
- “Demography and Health”, SONAR 2019
- “Cancer treatment revolution? Liquid biopsy and immunotherapy”, SONAR 2017
- “Ageing in dignity – the pivotal role of re/insurance”, SONAR 2017
- “Citizen science – an opportunity for insurance?”, SONAR 2017
- “Demographic inclusion at the workplace”, SONAR 2014

<sup>4</sup> “Health sector challenges and policies in the context of ageing populations \*”, UN’s Department of Economic and Social Affairs, November 2021.

<sup>5</sup> “FDA Grants Accelerated Approval for Alzheimer’s Disease Treatment”, 6 January 2023.

<sup>6</sup> “FDA approves new treatment for osteoporosis in postmenopausal women at high risk of fracture”, 9 April 2019.

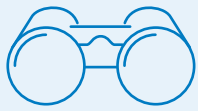
<sup>7</sup> Renée Mirkes, “Transhumanist Medicine: Can We Direct Its Power to the Service of Human Dignity?”, National Institute of Health, 29 March 2019.

<sup>8</sup> World Health Organization Ageing and health (who.int)

<sup>9</sup> “New Care Models: How insurers can rise to the challenge of older and sicker societies”, The Geneva Association, 18 August 2021.







# Overlooked exposures: emerging risk drivers for Life & Health insurers

Life & Health (L&H) business is about more than mortality and morbidity forecasts.

Morbidity expectations underpin health insurance, while life and longevity insurance solutions are based on mortality forecasts. COVID-19 was a reminder, however, that morbidity and mortality forecasts alone may not suffice for L&H sector predictions of the future.

The graphic below shows the emerging risk signals and trends that we have cited in previous SONAR reports as having an impact on L&H insurance, positive and negative. For instance, developments in science and technology can lead to healthier and longer lives. Other factors, like habits leading to obesity or lifestyle drugs taken for non-medical reasons, may do the opposite. Some developments can have both positive and negative impacts.

## Key takeaways

Many emerging risks in L&H are unrelated to epidemiological or life-science developments. Other important risk drivers often not taken into account include the reliability of energy and clean water supplies. These become especially important when a well-functioning healthcare system is needed most: in times of crisis.

For example, heat waves are expected to occur more frequently as temperatures warm. People who work outside, those with pre-existing medical conditions, the elderly, children and pregnant women are most likely to suffer adverse health effects in high temperatures, especially if they do not have access to mitigation measures. To furnish these, healthcare facilities need energy to operate and for cooling. In addition, patients need water for rehydration. As the number of heat days increases, patient admissions tend to rise. This adds to stresses on healthcare systems including via drinking water and electricity supplies, the latter because thermal and hydropower facilities need water to function.

Actuarial insurance models predict morbidity and mortality based on historical health data, and under the assumption that healthcare systems are well functioning given

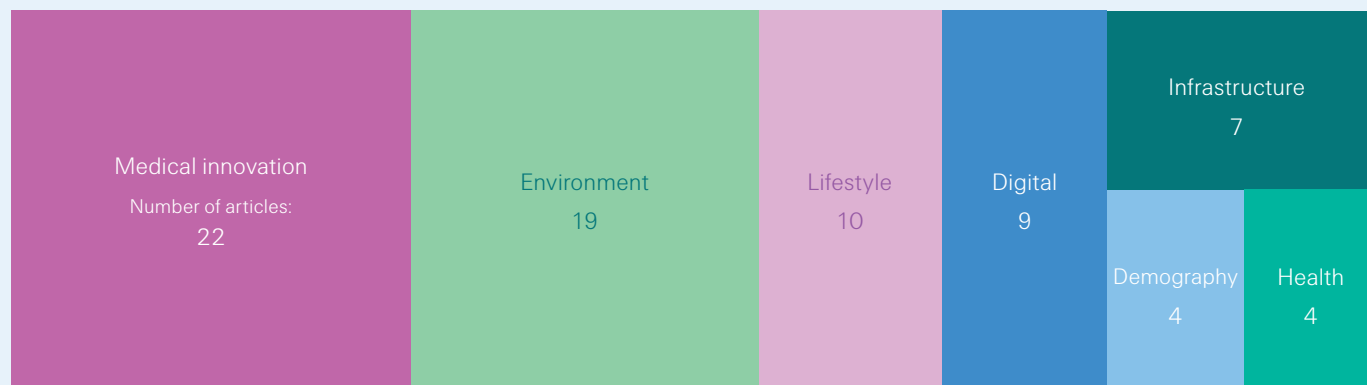
adequate availability of water and power. However, if in the future water or power shortages curtail the operation of healthcare services, actual outcomes may deviate considerably from the expected, especially if disruption persists for longer time periods.

Many other risk drivers have been on our radar for some time, also. For example, as populations age, the number of people vulnerable to communicable diseases increases. At the same time, however, some key issues like antibiotic resistance or underinvestment in vaccines remain unresolved. Such risk factors can compound the risks the insurance industry is already exposed to, such as increased morbidity and mortality in the case of epidemics/pandemics.

Over the last 10 years, two long-term global trends are especially prominent: climate change and the destruction of nature. Both bring their own health-related hazards while simultaneously reinforcing one another. Human-wildlife interactions increase the risk for zoonotic diseases, which in turn may result in epidemics and pandemics. Meanwhile climate change can exacerbate the range or spread of diseases. This may expose populations to communicable diseases that they have not experienced before, impacting morbidity and mortality rates in unexpected ways.

Actuarial modelling should adapt to account for both the positive (eg, in the field of medical innovation) and negative factors that could influence morbidity and mortality trends. The L&H industry should look beyond the traditional focus of health-related data to address new risk drivers. For instance, climate change modelling or risk mitigation for infrastructure related business interruption are already topics in non-life insurance segments for decades. Such insights can help in the task ahead: having a fresh look at scenarios and how to adapt to the qualitative risk changes the world is going through.

## SONAR 2013–2022 L&amp;H signals and trends



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# Bird flu – wake-up call for outbreaks of animal diseases?

The recent outbreak of Avian influenza is the largest ever and has killed millions of birds globally.<sup>10</sup> Transmission of the H5N1 virus from birds to other animals and (even) humans can trigger a range of insurance policies.

## Potential impacts

- In more mature markets, producers affected by government slaughter orders typically receive compensation. Payment may come from a compulsory livestock epidemic fund financed with insurance levies from livestock holders and/or government contributions, or directly from state budgets.
- Private insurance can help cover the difference (if any) between state-sponsored cash compensation and the market value of lost livestock and/or business interruption costs.
- There can also be workers' compensation claims, for farm workers infected with the virus.
- There can be claims and losses from agents in the overall food distribution chain due to shortages, in the case of birds, supply of eggs and poultry.
- Virus migration to humans at large scale (which is theoretically possible but has not yet happened) could impact mortality and morbidity covers.
- A learning from COVID-19 is that an influenza epidemic/pandemic increases interest in L&H cover. The affordability for such insurance remains a concern in many parts of the world.
- In case of an extreme pandemic or epidemic, travel restrictions and lockdowns may kick in, leading to supply chain interruptions and financial market impacts.

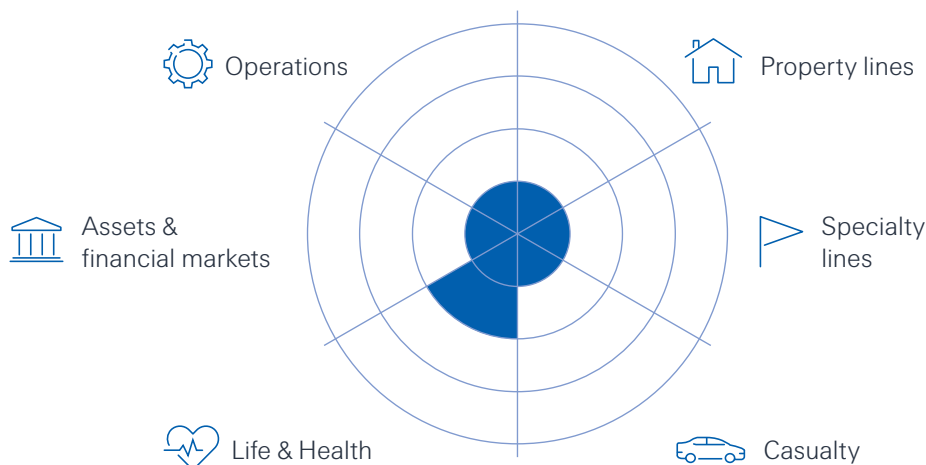
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



Last year at least 22 million chickens were culled in the US in an attempt to stop the latest outbreak of Avian flu. In total, more than 43 million hens were killed,<sup>11</sup> and many workers in the sector lost their jobs. In Japan, more than 10 million birds were culled, with over 90% being egg-laying hens. It was the worst outbreak of bird flu Japan has ever experienced.<sup>12</sup> Culling has a short-term effect of limiting the spread of the virus. But culling is ineffective as a long-term strategy because by killing infected birds, host resistance is also eliminated.

## Flying into humans

H5N1 was first reported in humans in Hong Kong in 1996 and there have been repeated cases since. In the winter of 2022/23, the World Health Organization (WHO) reported a few cases of transmission of bird flu to humans, with further cases reported in the media in the first quarter of 2023.<sup>13</sup> As of yet, there have been no reported cases of human-to-human transmission but, as the WHO said "we cannot assume that will remain the case, and must prepare for any change in the status quo".<sup>14</sup>



It is high impossible to prevent contact between wild animals and infected birds. The primary mode of virus transmission is when animals eat infected wild birds.<sup>15</sup> The bird flu virus has been detected in various mammals including foxes, sea lions, otters, black bears and pumas.<sup>16</sup> Recently researchers found minks in Spain infected with H5N1, suggesting that the virus can also spread from mammal to mammal.<sup>17</sup> There are vaccines for birds and more are under development<sup>18</sup> but realistically, only farmed birds can be vaccinated. Almost all wild creatures are non-vaccinated.

### The next pandemic?

Zoonotic diseases (pathogen spillover from animals to humans) have been a global health concern for several years. There have been worries that the recent outbreak of bird flu could become the next pandemic. Some factors escalating the risk are new variants of the virus, new mammal species being infected, climate change and shifts in bird migration routes.<sup>19</sup> The WHO recommends that countries strengthen surveillance in settings where humans and farmed or wild animals interact.<sup>20</sup> Avian flu is listed as a World Organisation for Animal Health disease, and national veterinary authorities must report outbreaks when certain thresholds are met.<sup>21</sup>

COVID-19 highlighted the weaknesses and gaps in the world's collective defences against pandemics, notably funding shortages and lack of a globally coordinated response to disease outbreak. A World Bank/WHO report speculates that the costs of the next pandemic will likely exceed those of COVID-19 if these weaknesses are not addressed.<sup>22</sup> A concern is pandemic fatigue, which could lead to a slower response to signs of a new pandemic, and

potential delays in executing the necessary mitigation actions.<sup>23</sup> Insurers would be well advised to translate the learnings from COVID-19 to reflect the possibility of virus crossover from animals to humans.

### Related articles in SONAR

- “How to better prepare for the next pandemic”, SONAR 2022
- “Vaccination – a shot worth more than politics and profitability”, SONAR 2019
- “Rising pandemic risk”, SONAR 2015
- “Emerging infectious diseases”, SONAR 2013

<sup>10</sup> “2022-2023 Confirmations of Highly Pathogenic Avian Influenza in Commercial and Backyard Flocks”, USDA, 18 January 2023, “Record avian flu outbreak sees 48m birds culled in UK and EU,” BBC News, 3 October 2022.

<sup>11</sup> “US egg factory roasts alive 5.3m chickens in avian flu cull – then fires almost every worker”, The Guardian, 28 April 2022, “Avian influenza outbreaks reduced egg production, driving prices to record highs in 2022”, The USDA, 11 January 2023.

<sup>12</sup> “Record 10 Million Birds Culled in Japan Due to Avian Flu”, Nippon, 24 January 2023.

<sup>13</sup> “Influenza at the human-animal interface summary and assessment, 5 January 2023”, The WHO, 5 January 2023., “Girl who died of bird flu did not have widely-circulating variant”, Nature, 28 February 2023., “China reports a third human case of the H3N8 strain of bird flu”, South China Morning Post, March 29 2023., “Chile detects first case of bird flu in a human”, Reuters, 30 March 2023.

<sup>14</sup> “WHO Director-General’s opening remarks at the media briefing – 8 February 2023”, The WHO, 8 February 2023.

<sup>15</sup> “Bird Flu in Pets and Other Animals”, The CDC, 22 February 2023.

<sup>16</sup> Avian Influenza, The WOA, March 2023.

<sup>17</sup> Agüero, M. et al. “Highly pathogenic avian influenza A(H5N1) virus infection in farmed minks, Spain, October 2022”, Eurosurveillance.org, Volume 28, Issue 3, 19 January 2023.

<sup>18</sup> “U.S. to test shots against bird flu outbreak, as Biden administration weighs poultry vaccinations”, CBS News, February 9 2023., “Could avian flu cause the next human pandemic?”, The Economist, 15 February 2023.

<sup>19</sup> “Avian Influenza”, WOA, March 2023, C.J. Carlson et al. “Climate change increases cross-species viral transmission risk”, Nature, 22 April 2022.

<sup>20</sup> “WHO Director-General’s opening remarks at the media briefing – 8 February 2023”, The WHO, 8 February 2023.

<sup>21</sup> “Avian Influenza”, The WOA, March 2023.

<sup>22</sup> “Analysis of Pandemic Preparedness and Response (PPR) architecture, financing needs, gaps and mechanisms”, The WHO and the World Bank, 22 March 2022.

<sup>23</sup> “Could avian flu cause the next human pandemic?”, The Economist, 15 February 2023., “Bird flu outbreak is a wake-up call for agriculture”, The Straits Times, 9 February 2023.











# Political and economic environment

# The global debt distress – fallout from rising interest rates

Interest rate hikes come at a time when public debt levels are at historic highs. Rising debt servicing obligations increase the risk of repayment defaults, and may constrict future public investments and economic growth.

## Potential impacts

- The rising cost of debt can call into question government finances and expose latent vulnerabilities built up among financial markets participants that leveraged up in the low-yield, low-volatility interest rate regime.
- High sovereign debt burdens could lead to reduced investment in infrastructure and public services, leading to potentially more failures in critical infrastructure and healthcare services.
- A decline in growth-oriented public investment could lead to higher unemployment, falling productivity and, potentially, social unrest. The latter could trigger credit and surety (C&S), property, business interruption (BI) and political violence (PV) covers.
- In parallel, the current cost of living crisis could exacerbate the debt crisis by weakening consumer purchasing power, leading to a decline in insurance purchases and ultimately diminishing societal resilience.

## Impact

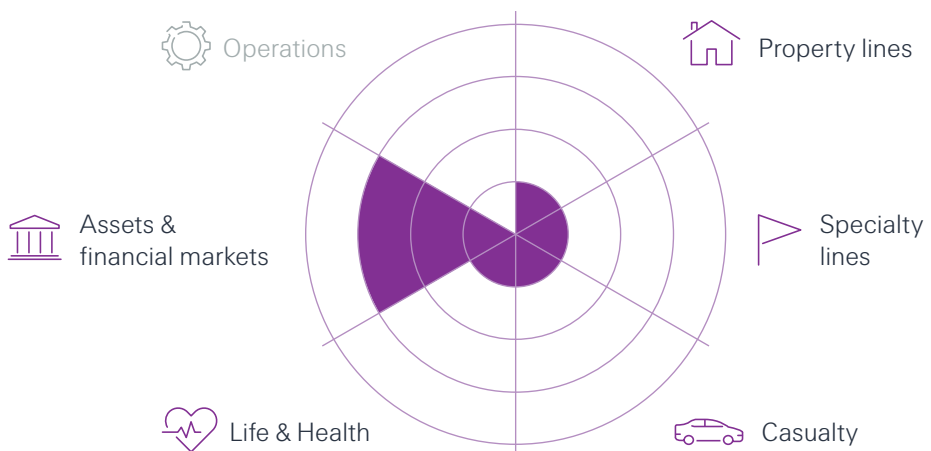


## Time frame

0 – 3 years

> 3 years

## Impacted business areas



The era of cheap financing is over. After a decade of low interest rates, central banks are hiking rates to contain the rapidly rising inflation sparked by expansionary fiscal policy during the COVID-19 crisis and higher commodity prices as a fallout from the war in Ukraine. These rate hikes come at a time when public debt levels are at historic highs. The low-interest rate environment in place since the global financial crisis of 2008–09 had provided an opportunity for governments to raise their debt to record levels, which the emergency fiscal measures actioned during the pandemic only added to. With higher rates, the costs of debt servicing (paying interest on existing debt) and of securing new debt have increased substantially. The very large public debt bills remain

sensitive to further interest rate hikes, bringing into question the sustainability of sovereign debt worldwide.

## Emerging market debt crisis risks

Emerging markets are relatively more exposed to rising interest rates, due to higher macroeconomic vulnerability and shorter-dated debt stocks that need to be rolled over in an environment of higher capital costs.<sup>24</sup> Countries with a high proportion of US dollar-denominated debt are particularly vulnerable on account of dollar strength.

Another factor that makes emerging markets more vulnerable to rising debt costs is creditor diversity and opacity of term loans. Over the past decade, the range of creditors of public debt in emerging economies has



broadened. It now includes quasi-sovereign entities and also the private sector participants like commodity traders and producers.<sup>25</sup> In 2020, private-sector creditors accounted for nearly 62% of emerging markets' external debt, up from 43% in 2000.<sup>26</sup> Meanwhile, external debt of emerging markets as share of GDP grew from 23% in 2008, to 31% in 2021. This represents a shift towards arguably more expensive and risky sources of funds than in the past, such as from international private and commercial lenders.<sup>27</sup> With that shift, creditor companies globally may have become more exposed to a potential debt crisis in emerging markets, with contagion spreading systemic vulnerability across large banks and private creditors. Meanwhile, China has become a large bilateral creditor to many emerging markets. Where public debt restructuring is needed, this could further strain geopolitical dynamics, as debt may be leveraged to pull emerging economies into the growing polarisation between large powers.<sup>28</sup>

#### Rising debt costs threaten social resilience globally

In advanced markets – where public debt levels are also at historically high levels – rising debt servicing obligations may constrict future economic growth, as a larger share of GDP goes to debt servicing payments and less to new growth-oriented investment or anti-cyclical fiscal policies.<sup>29</sup> Low levels of investment, coupled with failing productivity in advanced markets could spell rising unemployment and potentially increased social unrest. As a result, credit and surety (C&S), property, business interruption (BI) and political violence (PV) covers may be triggered. A weakening in public health services investment could also reflect in poorer L&H outcomes.

The build-up of sovereign debt poses significant risks to global resilience, with over-indebted governments potentially unable to provide public goods and social safety nets.<sup>30</sup> This situation may be exacerbated by the ongoing cost of living crisis, as the rising cost of food and energy reduces

consumer purchasing power and prices some households out of risk protection solutions. Combined, these factors could imply that insurance affordability will be threatened at a time when societal resilience has already been undermined.<sup>31</sup> Under these conditions, the global economy is ill-prepared for a next shock event.

#### Related articles in SONAR

- “Zombie companies – sustained by COVID-19 support”, SONAR 2021
- “Downturn and recovery”, SONAR 2020
- “Tipping the scale? – Intergenerational imbalances on the rise”, SONAR 2020
- “Hamstrung?: The risks of heavy government debt”, SONAR 2018
- “Hijacked money – political risk of forced investments”, SONAR 2017
- “Emerging market crises 2.0”, SONAR 2016
- “Fossil fuel mismanagement”, SONAR 2015
- “The great monetary experiment”, SONAR 2015
- “Contagious emerging market crisis”, SONAR 2014
- “Short-termism of macro-policy measures”, SONAR 2014
- “Run-away inflation and surging bond yields”, SONAR 2013

<sup>24</sup> “The IMF’s Fiscal Monitor Update – Surprise inflation lowers DM debt levels, Ems not so much”, Goldman Sachs Economics Research, April 2023.

<sup>25</sup> SwissRe Sigma, 2023, “Economic stress reprises risk: global economic and insurance market outlook 2023/24”

<sup>26</sup> The Unfolding Sovereign Debt Crisis | Current History | University of California Press (ucpress.edu)

<sup>27</sup> Soaring debt burden jeopardizes recovery of least developed countries | UNCTAD

<sup>28</sup> WEF, 2023, Global Risks Report 2023.

<sup>29</sup> Romer, Christina D., and David H. Romer. 2019. “Fiscal Space and the Aftermath of Financial Crises: How It Matters and Why.” Brookings Papers on Economic Activity, Spring, 239-331.

<sup>30</sup> WDR 2022 Chapter 5 (worldbank.org)

<sup>31</sup> Swiss Re Institute research estimated that inflation increase in 2022 alone widened the global insurance protection gap by USD 55 billion, or about 3.8% of the 2021 total. Source: <https://www.swissre.com/dam/jcr:3f36e9da-fe0f-401d-8648-9a12770ffc0f/2022-june-sigma-resilience-index-en.pdf>



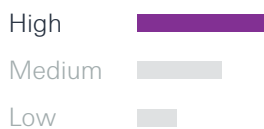
# Exclusive markets – when geopolitics polarises business

Free markets facilitate global re/insurance business. Further polarisation of geopolitical alliances could lead to mutually exclusive markets and impede global re/insurance businesses.

## Potential impacts

- Loss of access to a market on account of fragmentation of the global economy into regional blocs would have negative premium impacts. Protectionist measures could result in lower growth. Regulatory uncertainty affects long-tail business in particular.
- Complexity of geopolitical alliances and sanction regimes increases legal and compliance risk and costs for re/insurers.
- Operational challenges also include potential breakdowns of supply chains, threats of cyberattacks and commercial espionage.
- Liability and asset portfolios could come under stress in case of military conflict, and there could be strains on liquidity management, this increasing the cost of capital.

## Impact

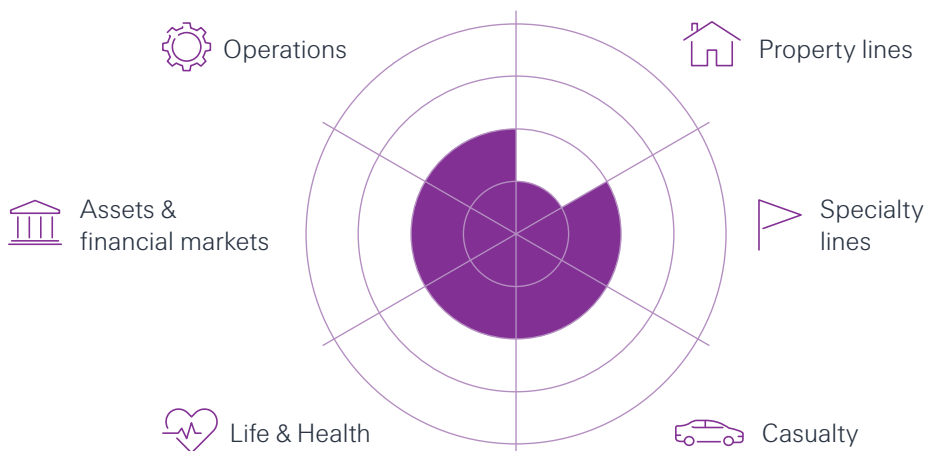


## Time frame

0 – 3 years

> 3 years

## Impacted business areas



The ability to move capital across borders underpins global re/insurance business by keeping the cost of capital low, thereby enabling re/insurers to help legal entities around the world strengthen their risk readiness. Currently, a number of developments threaten free market access and movement of capital, and also efforts to diversify insurance risk globally.

## Protectionist sentiment on the rise

A main benefit and ambition of globalisation has been the long-term reliability of legal and regulatory environments for domestic and international enterprises. Competition

among countries to attract business serves as a strong deterrent to overly politicise trade and investment relations.

However, things have changed. There has been an increase in trade barriers since the global financial crisis of 2008–09, and these now cover a wide gamut of goods and services.<sup>32</sup> In addition, governments have made defensive use of financial and technological dependencies to weaken competitors. An example are protectionist US policies like reshoring of chip manufacturing facilities to prevent China from gaining a decisive technological and strategic edge.<sup>33</sup> Though

driven by different motivations, western sanctions on Russia carry a similar idea: by cutting off access to the international economy, Russia's ability to wage war on Ukraine is undermined.

Also, secondary sanctions which first appeared in the 1990s are making a comeback. The seizure of assets of individuals without criminal conviction has made its way onto the political agenda. For instance, a current debate focuses on whether to confiscate the fortunes of wealthy Russians to compensate Ukraine for war damages.<sup>34</sup> Lastly, new payment systems have emerged, for example in India, Iran and Russia, making these countries less exposed to efforts by the west to close existing financing conduits.<sup>35</sup> And a few countries use crypto currencies to evade sanctions.<sup>36</sup>

### A new world order taking shape

These developments have seen a re-shaping of geopolitical alliances, which could potentially lead to a decoupling of polarised political blocs.<sup>37</sup> Three key developments signal the development of mutually exclusive economic blocs. The first is the growing popularity of economic sanctions in the case of conflict. The second is political realignment (notably in advanced markets) of the traditional left vs right divide, to nationalists vs globalists.<sup>38</sup> And the third development is the trend of friend-shoring, whereby firms relocate production operations to states with aligned values rather than set up in lowest-cost locations.<sup>39</sup> All three could impede the free movement of capital that allows businesses to operate "normally".

### Related articles in SONAR

- "Increased global uncertainty and risk awareness", SONAR 2022
- "Beggars thy neighbour? Global trade reordered", SONAR 2019
- "Globalisation, fragmented", SONAR 2019
- "A brave new world? – Emerging geopolitical risk", SONAR 2018
- "Hijacked money – political risk of forced investments", SONAR 2017
- "Island solutions – regulatory fragmentation", SONAR 2017
- "Reduced market access – protecting your own backyard", SONAR 2017
- "De-globalisation", SONAR 2015

<sup>32</sup> "Independent monitoring of policies that affect world commerce", Global Trade Alert, March 2023.

<sup>33</sup> S. Kreps and P. Timmers, Bringing economics back into EU and U.S. chips policy, Brookings TechSteam 20 December 2022.

<sup>34</sup> "Freeze – Don't Seize – Russian Assets", Foreign Policy, January 13 2023., "Could Switzerland seize Russian assets to rebuild Ukraine?", Swissinfo.ch, 22 June 2022.

<sup>35</sup> "Russia and Iran launch payment system as an alternative to Swift", Middleeasteye.net, 30 January 2023., "India's digital payments market will more than triple to US dollars 10 trillion by 2026: Report", India Times, 9 March 2023.

<sup>36</sup> "Cryptocurrencies and U.S. Sanctions Evasion: Implications for Russia", CSIS, 20 December 2022.

<sup>37</sup> In contrast to the Cold War, which was essentially a military-ideological competition with virtually no economic relations between the two blocs, today nation-state competition is multi-layered, encompassing military, ideological and technological facets, and also restrictions on trade and investments. As long as economic links between blocs remain, so too will opportunity to make defensive use of certain facets. Only with total decoupling, would this end.

<sup>38</sup> "Patriots vs globalists replaces the left-right divide", FT.com

<sup>39</sup> "The US 'friendshoring' experiment risks making enemies", FT.com, Braw, E. (2023). "Corporations Can't Ignore Geopolitics Anymore", Foreign Affairs, Jan 4.



# Climate change, trade and geopolitics – new risks in the Arctic

Climate change is driving the “Arctic opening”, giving rise to new shipping and trade routes. The economic and strategic potential of the region is growing. So too are the potential for environmental exposure and geopolitical frictions. These hamper cooperation and governance, creating new and emerging risks.

## Potential impacts

- Demand for insurance related to economic activities in the Arctic region – ranging from trade (marine), to fishing, to mining and to tourism, is likely to grow.
- Economic and geopolitical uncertainties bring strategic and reputational risks.
- Potential for business disruptions in the region, and also for secondary effects on global supply chains is elevated.
- Arctic opening and permafrost melt give rise to environmental risks, which could impact property and specialty lines of business. Pollution could trigger casualty insurance policies.
- The release and migration of pathogens formerly frozen in ice and permafrost could be the source of new epidemics/pandemics.
- Cooperation among the eight member states of the Arctic Council has been suspended, fostering governance, regulatory and legal uncertainties. These will impact also the regulatory framework for insureds and insurers.
- The prospect of military conflict in the region has increased, changing the risk landscape for entities active in the region and their insurers.

## Impact

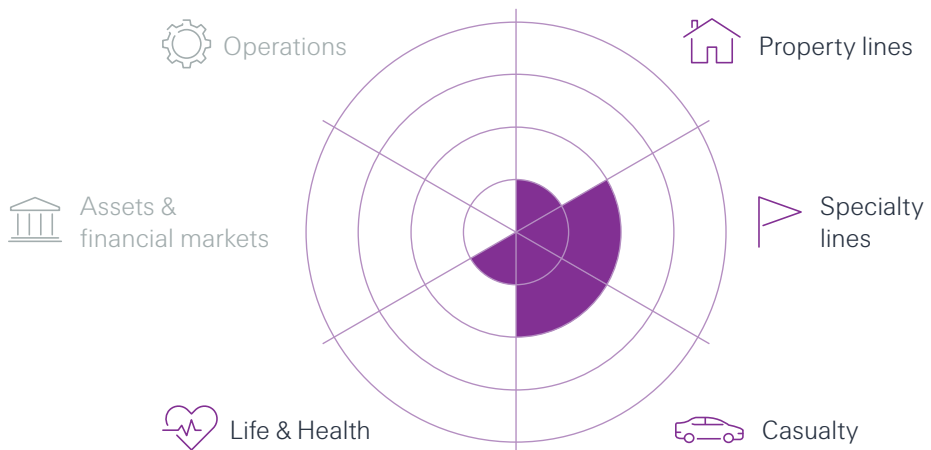


## Time frame

0 – 3 years

> 3 years

## Impacted business areas



## Climate change opens new Arctic routes

The Arctic – the ocean and adjacent land north of the northern polar circle (66°32'N)<sup>40</sup> – is warming two to three times faster than the rest of the globe.<sup>41</sup> With this, the Arctic sea-ice cover is melting, becoming both less permanent and more penetrable. This is opening up new, short-cut shipping routes and opportunities for tourism. Marketing for exclusive north pole cruises is already happening.<sup>42</sup> The Arctic opening also presents new areas for mining, oil and gas drilling, and fishing.

## Arctic risks

All told, the Arctic opening also comes with many challenges and uncertainties. For example, the economic competitiveness of arctic routes over traditional shipping routes is not certain.<sup>43</sup> And, with the rise of sustainability commitments and regulations, the value of mining and fossil fuel development in the region is also questionable.<sup>44</sup>

Fast environmental changes are also happening. Rising temperatures are releasing methane and other greenhouse gases formerly bound to the ground and ice, accelerating

global warming further. Melting sea ice covers and permafrost threaten the foundations of buildings in the region, and livelihoods. Arctic populations are seeing their sources of marine and wild food disappearing, and their homes being destroyed.

The risk profile of many activities in the region will change as temperatures continue to rise. Among the effects are sea-level rise, which threatens settlements and industrial installations. Also, toxic waste deposited in prior decades in the course of military, mining and other adventures can be uncovered and dispersed, leading to environmental pollution.<sup>45</sup> And, melting ice and thawing permafrost may result in release of infectious pathogens, new sources of potential epidemics and pandemics. Other threats are wildfires and heatwaves, and their effect on soil quality and nutrients. These also change the composition of the Arctic forests, which could render the latter less resilient.

### Strategic competition and governance gaps

The trend of increasing environmental and economic risk is paired with the strategic interests and military ambitions of the eight nations with coasts on the Arctic sea: Norway, Sweden, Finland, Russia, the US, Canada, Denmark and Iceland. For example, Russia sees Arctic ice melting as a defence threat.<sup>46</sup> Many nations have military presence and infrastructures in the region, and some are ramping up of said capacities, with Russia's invasion of Ukraine having raised the risk of conflict in the region. This coincides with NATO's integration of Finland (joined in April 2023) and Sweden (application pending).

The Arctic Council is the governance body for the region, representing the aforementioned eight nations. All Council decisions and statements require consensus of the eight member states.<sup>47</sup> Russia is chairing the Council in 2022–23. After its invasion of Ukraine, the other seven member states declared a joint suspension of collaboration with Russia in the Council.<sup>48</sup> Decision-making is therefore

on hold, which is in turn blocking joint efforts in research, environmental monitoring and the establishment of common standards for shipping and oil and gas development. The coinciding increases in economic interests, environmental change and geopolitical tensions make the Arctic a hotbed for emerging risks and potential risk accumulation.

### Related articles in SONAR

- “Thawing permafrost – accelerating climate, infrastructure and health hazards”, SONAR 2022
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- “Bugs on the march – underestimated infectious disease”, SONAR 2017
- “Blue ocean economy”, SONAR 2017
- “Seabed mining”, SONAR 2016
- “Collapse of oceanic ecosystems”, SONAR 2014
- “Methane hydrate – an upcoming energy source?”, SONAR 2014

<sup>40</sup> Definitions of the Arctic vary, depending on the measure applied. Through a climate zone lens, it is traditionally confined by the 10 degree Celsius July isotherm, an abstract line, north of which the mean temperature remains below 10 degrees even in the warmest month in a multi-year average. M. Paul and G. Swistek, *Russia in the Arctic*, Berlin 2022, p. 9.

<sup>41</sup> John P. Holdren, *Climate change is rapidly transforming the Arctic: Why everybody should care*, The Hill, 6 Sept. 2021.

<sup>42</sup> For an example see <https://en.ponant.com/destinations/north-pole>

<sup>43</sup> To navigate the geographically shorter northern passage ships need to be accompanied by icebreakers or to be equipped with icebreaking capability. This comes with higher fuel consumption also in open water. Competitiveness of Arctic shipping routes is thus depending on fuel prices and energy efficient ship design. New energy efficiency regulations for open water ships favor bow designs which are less effective in ice, thus putting a question mark behind arctic routes during cold seasons. Rob Hindley, *Future of Arctic Ships*, Aker Arctic 2022.

<sup>44</sup> Industrial development in the Arctic | GRID-Arendal ([grid.no](https://grid.no)).

<sup>45</sup> Matt Simon, *A toxic time bomb is ticking in the Arctic*, WIRED 4 April 2023.

<sup>46</sup> M. Paul and G. Swistek, *Russia in the Arctic*. SWP Research Paper 3, Berlin 2022.

<sup>47</sup> About the Arctic Council | Arctic Council ([arctic-council.org](https://arctic-council.org)).

<sup>48</sup> In June 2022, the governments of Canada, the Kingdom of Denmark, Finland, Iceland, Norway, Sweden, and the United States issued a joint statement to confirm limited cooperation – excluding projects involving participation of the Russian Federation: Joint Statement on Limited Resumption of Arctic Council Cooperation – United States Department of State



# Accidental or hostile – dangers from nuclear radiation release

Nuclear energy facilities can suffer damage from different sources: war, intentional interference by hostile parties, unintentional mishaps in daily operations or maintenance, and damage from catastrophic natural hazards. Radiation release from any such event can trigger insurance claims across many lines of business.

## Potential impacts

- Economic activity in the region of the radiation release could slow significantly or come to a standstill, potentially impacting financial markets globally.
- Even a small-scale radiation release event can trigger claims for property losses.
- Radiation events can lead to supply chain disruptions and contingent business interruption.
- Human exposure to radiation could trigger claims in mortality and morbidity in L&H insurance on account of sudden death and illnesses that people subsequently suffer.
- A release event may lead to shortages of food and other life necessities (eg, clean water and energy).

## Impact

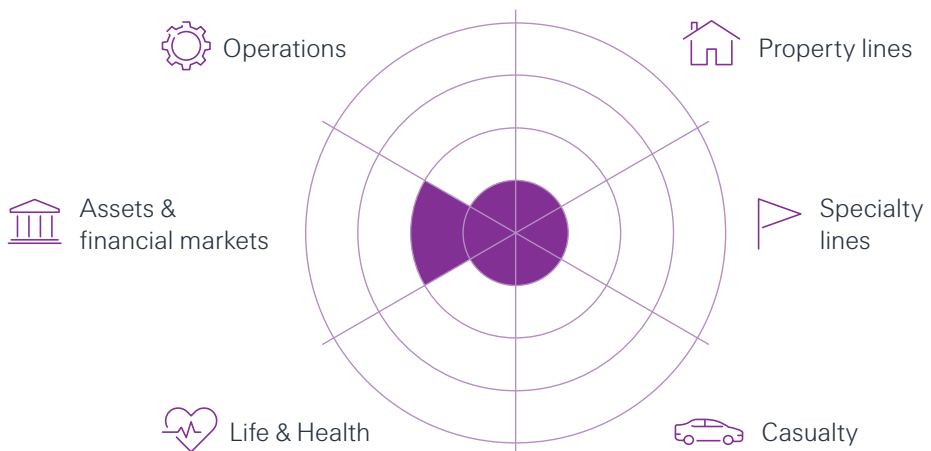


## Time frame

0 – 3 years

> 3 years

## Impacted business areas



Radiation release from a nuclear energy facility is an ever-present threat. Additionally, radiation release could result from deployment of nuclear weapons in a wartime, the most recent example being the war in Ukraine, and threats from Russia to use nuclear weapons. In addition, damage from fighting and shelling near nuclear plants in Ukraine (eg, Zaporizhzhia) could damage safety systems and confinement structures.

## Rogue actors vs “innocent” sources of danger

So-called “rogue nations” also pose a geopolitical threat. North Korea is a nuclear power, and regularly test-launches ballistic missiles.<sup>49</sup> And it is widely assumed that Iran is

pursuing nuclear armament.<sup>50</sup> Outside of the context of tension and war, nuclear systems can be targets of intentional sabotage by a range of non-state hostile parties, such as cybercrime gangs and terrorist organisations.<sup>51</sup>

Unintentional mishaps in daily operations or maintenance can also result in radiation release. Accidents related to the nuclear industry extend beyond those related to preparing fuels for power plants and disposing of the waste. A recent case was the loss of a capsule of radioactive material during transport in Australia.<sup>52</sup> Many other incidents can occur outside a nuclear energy facility.<sup>53</sup> Nuclear facilities can also suffer catastrophic damage if struck by a natural disaster like an earthquake or tsunami.<sup>54</sup>



### Potential claims across multiple lines of business

In case of a nuclear radiation release event, a series of possible loss scenarios may occur. Property damage, a surge in cancer cases, and disruptions to food production and industrial supply chains are among potential insurance claim outcomes. If a nuclear release incident occurs close to a location that is a financial centre, a standstill in economic activity could have ripple effects on financial markets globally. All these may result from damage to a nuclear power plant or processing facility by any of the above-mentioned sources of threat. While war is typically excluded from insurance covers, indirect consequences from deployment of nuclear weapons may nevertheless trigger policies if radiation fallout crosses international borders.

### Related articles in SONAR

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- “Insatiable demand for energy: Infrastructure in transition”, SONAR 2018
- “Decaying infrastructure”, SONAR 2015
- “Super nat cats”, SONAR 2015
- “Underestimated nat cat exposure”, SONAR 2013

<sup>49</sup> “After record year of arms tests, what’s in North Korea’s arsenal?”, Al Jazeera, 9 November 2022.

<sup>50</sup> Iran and Nuclear Weapons Production, Congressional Research Service, 14 April 2023.

<sup>51</sup> “Addressing Cyber-Nuclear Security Threats”, NTI, 2022.

<sup>52</sup> “Staggering recovery of tiny radioactive capsule in WA still leaves many questions unanswered”, The ABC, 1 February 2023.

<sup>53</sup> “Major Nuclear Reactor Incidents”, Atomic Archive (atomicarchive.com). Accessed May 2023.

<sup>54</sup> “Protecting Nuclear Power Plants Against Nature’s Fury”, IAEA, 23 February 2010.











# Technological and natural environment



# Hooked online – anxiety and distraction from social media binging

Attention-grabbing online platforms weigh on peoples' ability to concentrate and are being linked to poor mental health outcomes. The companies profiting from the attention economy may face increasing liabilities in the future.

## Potential impacts

- Overuse of digital platforms is associated with depression, anxiety, sleeping problems and others. This may result in long-term mental health issues, leading to disability and, in severe cases, mortality claims.
- Mental illness and a more sedentary lifestyle come with co-morbidities such as obesity and cardiovascular diseases.<sup>55</sup>
- Overuse of digital platforms can be distracting and lead to omissions in safety-critical functions (eg. system steering roles, security guards, etc) or accidents. This could impact liability lines of business: motor liability (car accidents), general liability, professional indemnity and employer liability/workers' compensation (workplace accidents) insurance.
- Increasingly, the companies behind digital platforms may be held liable for negative effects experienced by users, creating a new risk landscape for them.

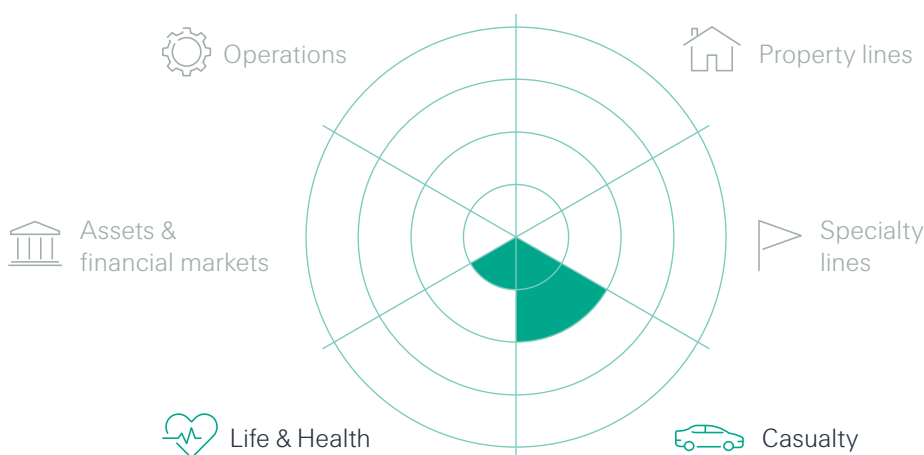
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



Social media, dating, streaming, gaming and other online platforms have become part of daily life. The business model of these platforms is based on constant hyperstimulation and novelty.<sup>56</sup> They drive profits by maximizing user engagement and time spent online. But does constant interaction with online platforms affect a person's brain, mood and mental health?

## Too much stimulation?

The internet exposes users to an infinite stream of media, packed with notifications and hyperlinks, and directions to engage with multiple inputs simultaneously.<sup>57</sup> Research

suggests that increased media multi-tasking lessens our ability to perform tasks that need sustained attention.<sup>58</sup> Evidence is also emerging of narrower attention scope right after spending time online.<sup>59,60</sup>

Online platforms try to keep users scrolling, swiping and viewing for as long as possible. On many platforms, algorithms generate bite-sized content targeting the reward system of the human brain, enticing the viewer and making it difficult to look away. This encourages passive and addictive behaviours, such as doom scrolling (excessive scrolling through negative news) and/or binge-watching

(watching online videos and streamed content for long periods of time). Such behaviours have been associated with increased levels of stress, anxiety, depression and sleep problems.<sup>61</sup>

### Who's responsible?

Awareness of the risks associated with excessive use of digital platforms is on the rise, and tools such as screen time control apps have become more popular. Even so, specific content on platforms can contribute to poor mental health. For example, in 2022 multiple lawsuits were filed against TikTok and Instagram on product liability grounds. The assertion was that algorithms generating content about dieting fostered teen eating disorders.<sup>62</sup> In addition, the longer-term impacts of online stimulation are uncertain. For instance, the effect on brain development and cognitive function in children is not well understood, yet. Today, what is deemed "normal" use of online platforms may be a breeding ground for longer-term mental health issues, in turn leading to more household spending on health and higher associated insurance claims.

There is a view that the companies profiting from our attention should hold more liability. The Digital Services Act (DSA) currently pending final approval by member states of the European Union (EU) mandates that platform operators develop codes of conduct and provide transparency on their algorithms.<sup>63</sup> This may lead to a new liability risk landscape for companies active in the field.

In the US, Section 230 of the Telecoms Act of 1996 has so far protected digital companies from liability around third-party content.<sup>64</sup> The pending case of Gonzalez vs Google will show whether this can be applied to algorithm-based

recommendations on digital platforms as well.<sup>65</sup> The outcome may well shape if and how digital companies are held liable for the content on their platforms and the moderation thereof.

### Related articles in SONAR

- "Out of sight, out of mind – mental health issues among the young", SONAR 2020
- "Eyes wide shut – the world is sleeping less", SONAR 2018
- "Dangerous games – risks of e-sports", SONAR 2017
- "The human factor – stress and fatigue in safety-relevant jobs", SONAR 2017
- "Digital slander", SONAR 2014

<sup>61</sup> E.G. Wilmoth et al, Sedentary time in adults and the association with diabetes, cardiovascular disease and death: systematic review and meta-analysis *Diabetologia* 55 (2012), p 2895–2905; A. Chaddah et al., Mental health and cardiovascular disease, *American Journal of Medicine* 129 (Nov. 2016), p 1145–1148.

<sup>62</sup> Online platforms & mental health – a policy proposal ((Harvard Kennedy School, CARR CENTER for Human Rights Policy) (2022)

<sup>63</sup> K. Kee Loh et al., "How Has the Internet Reshaped Human Cognition?", Volume 22, Issue 5, *Sage Journals*, July 9 2016.

<sup>64</sup> M. Peng et al., "Minds and brains of media multitaskers: Current findings and future directions", *PNAS*, 1 October 2018.

<sup>65</sup> "Attentional scope is reduced by Internet use: A behavior and ERP study", *PLOS One*, June 8 2018.

<sup>66</sup> J. Firth et al., "The 'online brain': how the Internet may be changing our cognition", *Wiley Online Library*, May 6 2019.

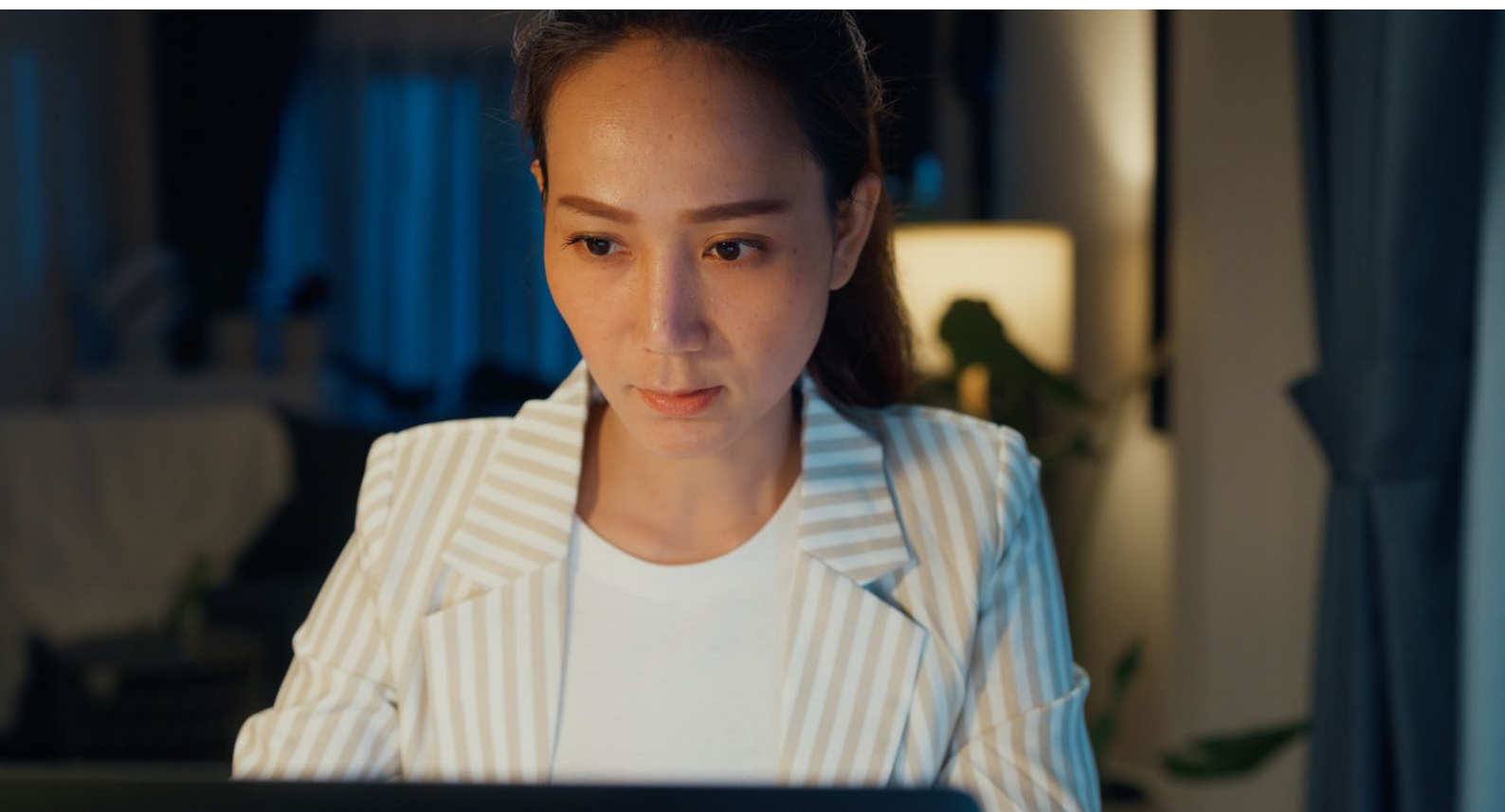
<sup>67</sup> Z. Alimoradi et al., "Binge-Watching and Mental Health Problems: A Systematic Review and Meta-Analysis", *MDPI*, June 20 2022; B. McLaughlin et al., "Caught in a Dangerous World: Problematic News Consumption and Its Relationship to Mental and Physical Ill-Being", *Taylor & Francis Online*, 23 August 2022.

<sup>68</sup> "Social media is a defective product, lawsuit contends", *Politico*, 26 January 2023. "Two new lawsuits claim Meta's Instagram caused eating disorders", *Reuters*, 26 July 2022.

<sup>69</sup> P. Popiel, "The Push to Regulate Digital Markets and Services", *Current History*, January 2023.

<sup>70</sup> See Telecommunications Act of 1996, Federal Communications Commission.

<sup>71</sup> Google v. Gonzalez LLC, *American Civil Liberties Union*, 27 February 2023.





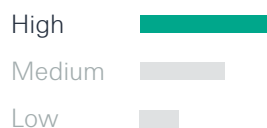
# AI getting hacked – systemic vulnerabilities of a booming technology

The growing use and capabilities of Machine Learning (ML) systems come with new vulnerabilities. Insufficient awareness and governance will expose increasing numbers of services to new attacks.

## Potential impacts

- The widespread use of AI and its systemic vulnerabilities raise concerns that adversarial ML could lead to risk/loss accumulation.
- Model evasion could impact automated insurance claims processing and facilitate fraud.
- Fraudsters might target the ML systems used for insurance distribution and pricing, leading to unwanted shifts in risk exposure.
- Professional liability lines: ML system failures or data breaches could trigger software producers' or distributors' professional indemnity and errors and omissions (potentially also director and officers') claims.
- Public reports of hacks/adversarial ML attacks could cause reputational damage.
- Targeted data poisoning could lead to unexpected high failure rates and, for instance, in case of autonomous cars or medical diagnosis software, trigger casualty or health claims.

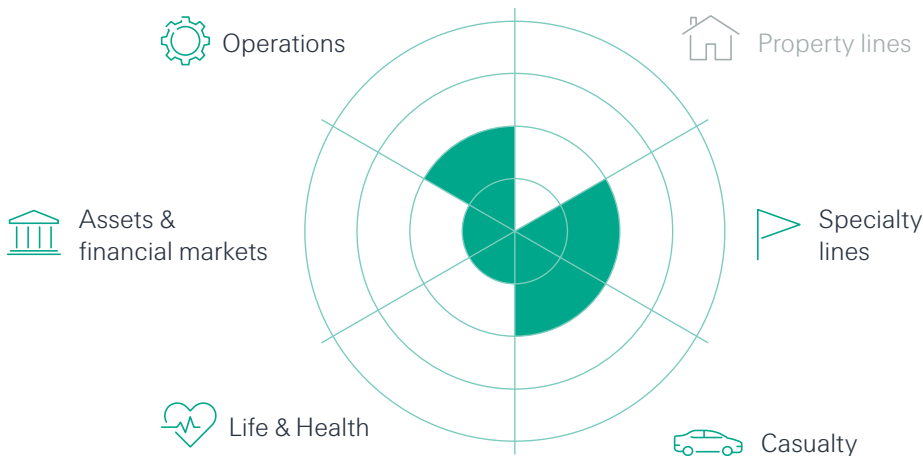
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas

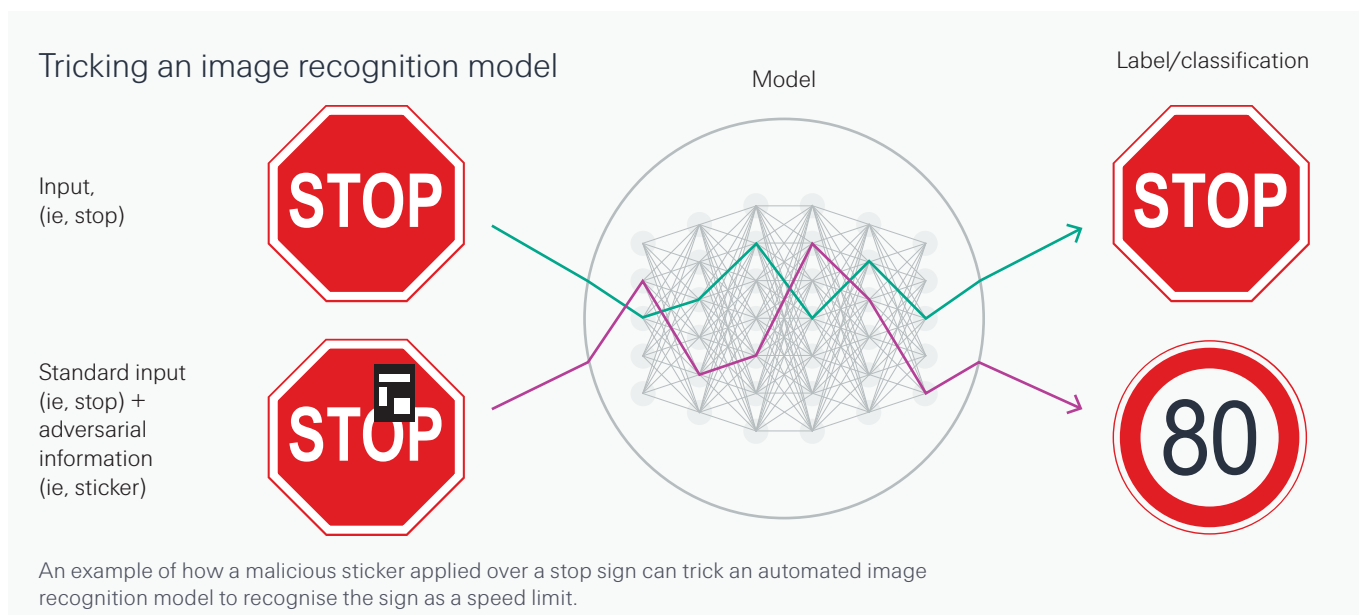


The use of Machine Learning (ML) systems is increasing rapidly. However, practitioners often do not understand, think about, or protect against an emerging class of vulnerabilities: “adversarial ML”. This threat encompasses any targeted exploitation or hacking of AI systems, leveraging ML-specific vulnerabilities.<sup>66</sup>

Together with recent progress in Deep Learning,<sup>67</sup> research interest in adversarial ML has increased notably.<sup>68</sup> Professional hackers are not only able to trick models into making mistakes or leaking information. They can also harm model performance by corrupting training data and/or stealing and extracting ML models.

The following examples showcase the diversity of potential ML vulnerabilities:<sup>69</sup>

- 1. Data/model poisoning (backdoors):** A backdoor responds to a trigger (specific data pattern) and produces a pre-determined outcome (eg, an exceptionally high creditworthiness or insurance score). Backdoors can be introduced using malicious pre-trained models (used as a basis for specialized ML) or malicious data, for instance by a disgruntled employee.
- 2. Model evasion:** Attackers could use adversarial ML to produce patterns that mislead ML systems and are difficult for humans to notice.<sup>70</sup> For example, a visual



pattern that – if applied to a car as a sticker – leads an automated motor claims tool to misjudge damage (see infographic for an example of a malicious attack on a stop sign).

**3. Membership inference** attacks can leak sensitive information used for the original training of ML models (eg, a confidential dataset that was used during a system's learning phase). By constructing a series of targeted queries, attackers can extract training data points with high probability. This raises concerns over data protection issues.

### Systemic impacts

The use of complex machine learning systems is becoming more widespread. By design their outputs are difficult to check for mistakes (both by humans and other ML). Furthermore, the systems are inherently vulnerable to adversarial attacks.<sup>71</sup> These vulnerabilities pose risks to insurers and businesses alike, and are a challenge for regulators also.<sup>72</sup>

Regarding risks relevant for insurers (beyond cyber insurance), there are increasing opportunities for fraud, as well as potential claims in professional indemnity and errors and omissions (E&O) lines for ML failures and data breaches. Furthermore, adversarial ML attacks leaking to the media (however small the impact might be) could directly impact the reputation of insurers and/or their assets. Model-stealing could lead to intellectual property (IP) loss. Beyond that, data leakage or non-compliance with current and upcoming data and AI regulation<sup>73</sup> might trigger fines. In some instances ML malfunction could cause harm or accidents (autonomous car crashes, medical misdiagnosis<sup>74</sup> etc) triggering casualty or health covers.

### Mitigations

Strict access management, suitable usage limits and data governance can go a long way in reducing attack surface

(areas of vulnerability that can be attacked). While not always applicable, simply not exposing models to the internet, and using only trusted (eg, not automatically collected) data, are powerful risk mitigation strategies. Getting all this right is by no means an easy feat, however. It entails making security and data governance a core feature of ML development and deployment, and striking a balance between usability and privacy/IP-protection. Starting this transition now will make ML applications more resilient in the future.

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- “Algorithms are only human too – opaque, biased, misled”, SONAR 2018
- “Blame your robot – emerging artificial intelligence (AI) legislation”, SONAR 2017

<sup>66</sup> This article focuses on malicious actors. Broader algorithmic risks have been covered in “Algorithms are only human too – opaque, biased, misled”, SONAR 2018.

<sup>67</sup> Deep Learning refers to ML using deep artificial neural networks (networks with many layers/parameters).

<sup>68</sup> See Biggio, B. and F. Roli, “Wild patterns: Ten years after the rise of adversarial machine learning”, Pattern Recognition (317–331), 2018; for a history of Adversarial ML.

<sup>69</sup> The spectrum of possible attacks is much broader than the presented selection. See for example, MITRE's ATLAS™ (Adversarial Threat Landscape for Artificial-Intelligence Systems) or “Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations”, NIST, 2023.

<sup>70</sup> In 2018 two Chinese citizens hacked into government's tax system by tricking facial recognition and stole US dollars 77 million. Cf “Faces Are the Next Target for Fraudsters”, Wall Street Journal, 2021.

<sup>71</sup> Cf eg Tramer, F. et al., “On Adaptive Attacks to Adversarial Example Defenses”, Advances in Neural Information Processing Systems (NeurIPS) 33, 2020.

<sup>72</sup> Eg “AI Risk Management Framework”, NIST, 2023; or “Recommendation of the Council on Artificial Intelligence”, OECD, 2022; or “Artificial intelligence governance principles”, EIOPA, 2021; or “Adversarial Machine Learning and the Future Hybrid Battlespace”, NATO, 2021. The KPMG “Cyber Trust Insight”, 2022 quotes Microsoft Vice President Ann Johnson: “We're doing a lot of work on adversarial AI because we believe that will be the next wave of attack.”

<sup>73</sup> Such as the GDPR, the upcoming EU AI act (cf “How AI is transforming governance and risk management in insurance”, EY Denmark, 2022), and the Algorithmic Accountability Act in the United States.

<sup>74</sup> Cf Finlayson, S., et al., “Adversarial attack on medical machine learning”, Science, 2019.



# Solar radiation management – risks from reversing climate change

Reflecting the sun's energy to address climate change effects is gaining more attention as the impacts of global warming become more visible. But the process, known as solar radiation management comes with many risks.

## Potential impacts

- Manipulating climate via SRM can lead to extreme weather events including flood, drought and windstorms, these triggering property losses and other claims.
- The release of compounds into the atmosphere impacts many countries. The nation which deploys SRM may not be the one that suffers harmful side effects. Deciding which legal frameworks and

governance for negative side effects to apply will be important.

- Atmospheric climate engineering may shift the zones for communicable diseases, heat stress and extreme weather events, which can trigger new mortality and morbidity patterns. Such shifts can impact L&H business.

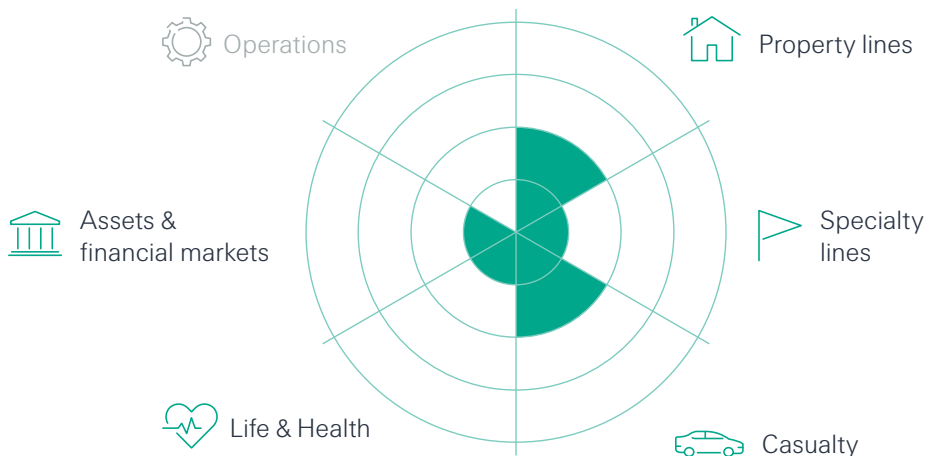
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



Solar radiation management (SRM) is an umbrella term for approaches to reduce global warming by intentionally reflecting increasing amounts of incoming sunlight back to space. SRM does not impact the root-cause of global warming, namely greenhouse gas (GHG) emissions. Its utility is to lower temperatures. However, if implemented and then suddenly terminated, the result of SRM could be fast temperature increases, bringing on related weather effects.<sup>75</sup>

SRM approaches include Cirrus Cloud Thinning (CCT), Ground-based Albedo Modification (GBAM), Marine Cloud Brightening (MCB) and space-based approaches involving

placing mirrors, shades or reflecting particles in the space between the sun and the earth.

To date, one of the most-studied SRM options is the stratospheric aerosol injection (SAI). SAI is the intentional release of highly reflective fine particles or their precursors at an altitude of 20–25 km. The effect would be similar to the Mount Pinatubo eruption on 15 June 2001. The volcano threw about 15 million tons of sulphur dioxide into the stratosphere, where it reacted with water to form a hazy layer of aerosol particles. Strong stratospheric winds spread these aerosol particles around the globe. Over the next 15 months, the average global temperature dropped by about 0.60°C.<sup>76</sup>

## The risks and benefits of SRM<sup>77</sup>

|                                                      | Impacts of SRM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | Positive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Negative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Impact of SRM on target risk                         | <b>Climatic benefits</b><br>SRM would reduce the future risks associated with temperature rise in most regions of the world. The most important benefits include: <ul style="list-style-type: none"> <li>■ reduced frequency and intensity of extreme temperature and precipitation events</li> <li>■ slowing sea level rise due to slowed melting of Arctic sea ice and mountain glaciers, and also the Greenland and Antarctic ice sheets</li> <li>■ reduced weakening of the Atlantic meridional overturning circulation<sup>78</sup></li> <li>■ reduction in the intensity of tropical cyclones</li> <li>■ slowing of soil moisture decline</li> <li>■ slight reduction in atmospheric carbon dioxide concentrations</li> </ul> | <b>Climatic risks</b><br>SRM does not reverse climate change. It is a different type of climate change in itself with distinct impacts, some of the most important of which include: <ul style="list-style-type: none"> <li>■ unintended warming or excessive cooling due to uncertainty in estimates of the amount of SRM needed</li> <li>■ changes in volumes of regional precipitation</li> </ul>                                                                                                                                                                                                                                             |
| Ancillary impacts of SRM: non-climate change impacts | <b>Co-benefits</b><br>SRM could have the side benefits of reducing other risks or adding value, both biophysical and social. Examples include: <ul style="list-style-type: none"> <li>■ reduced tropospheric ozone</li> <li>■ increased availability of water over land in the tropical regions</li> <li>■ reduced adverse health effects due to climate change</li> <li>■ equitable for rich and poor nations</li> <li>■ could help avoidance of climate tipping points</li> <li>■ relatively cheap to deploy</li> </ul>                                                                                                                                                                                                           | <b>Countervailing risks</b><br>Depending on the technology employed, SRM introduces some novel risks, both biophysical and social, including: <ul style="list-style-type: none"> <li>■ increased acid deposition in areas of high latitude</li> <li>■ effects on stratospheric ozone</li> <li>■ potential for international conflict and other societal risks because countries harmed may act against countries which deploy SRM</li> <li>■ potential interactions with a major volcanic eruption</li> <li>■ a sudden termination of SRM would likely bring climate/weather effects</li> <li>■ increase in salt deposition over land</li> </ul> |
|                                                      | <b>Positive or negative impacts</b> <ul style="list-style-type: none"> <li>■ influence on motivation for emissions abatement policy/behaviours</li> <li>■ light diffusion and dimming, and its effects on human health, ecosystems and agriculture</li> <li>■ effects on procedural and distributional justice, and other ethical concerns</li> </ul>                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

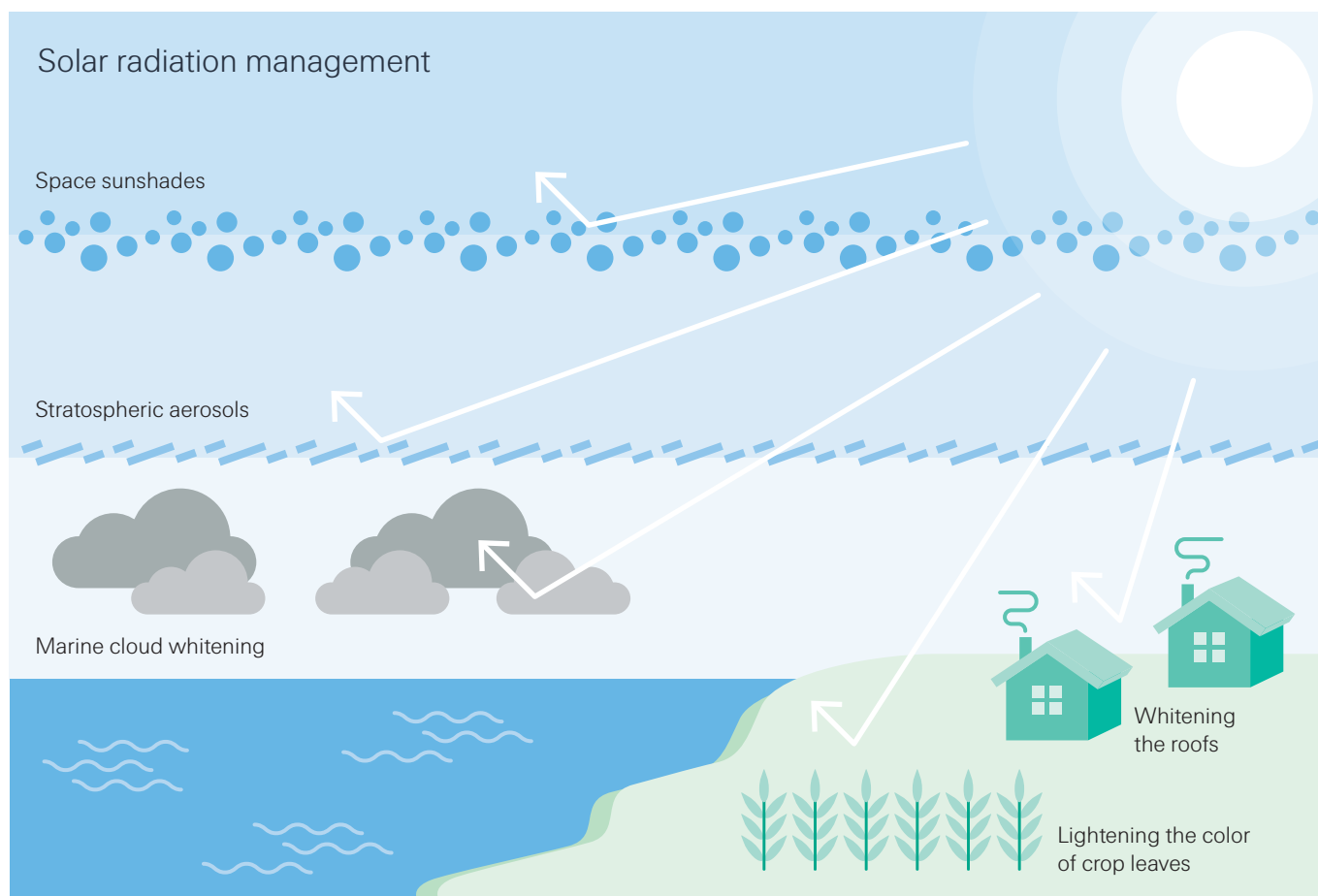
Source: Adapted from Solar radiation modification – A risk-risk analysis; Carnegie Climate Governance Initiative 2022

### Most important risks for the insurance industry

**Unintended climate changes.** SRM is essentially climate change in reverse: it cools the earth. Just as current temperature increases are likely fuelling changes in climate (drought, wet seasons) and extreme weather events (storms, flooding), so too will cooling via SRM. To predict

the effects of cooling, new models are needed reflecting where, what cooling agent is used and how much, and for how long it is injected into the stratosphere.<sup>79</sup> As models are currently under development, it is difficult to say which countries/regions would benefit or be harmed by consequent changes in climate and weather patterns. Any pattern changes, however, could lead to unanticipated losses for insurers.<sup>80</sup> One example of this could be a shift in





Schematic representation of selected Solar Radiation Management (SRM) geoengineering methods.

the geographical range of communicable diseases like malaria, which could redistribute in developing countries.<sup>81</sup> It could also lead to an increase or geographical shift of extreme weather events like droughts or hurricanes.<sup>82</sup> In general benefits and risks would most likely not be fairly distributed globally. Hence, the question is how to compensate those experiencing negative effects.

**Potential disruption of ecosystem services.**<sup>83</sup> SRM could change many climate variables other than temperature that are important for ecological systems. These include key ecological drivers, such as the ratio of diffuse to direct radiation, ultraviolet (UV) radiation, the connection between temperature and CO<sub>2</sub>, precipitation distribution, seasonality, acid precipitation, air quality and changes to surface ozone to name just a few. The resulting possible consequences of SRM implementation are wide ranging, from species decline and relocation, to populations stabilisation and even growth, changes to ecosystem processes and the emergence of novel ecological communities adapting to new climates. In turn, these could

trigger agricultural losses, health problems (ozone and UV radiation) and migration, bringing also insurance losses.<sup>84</sup>

**Potential for international conflict.** Some countries may pursue climate engineering while others condemn the practice. This would likely see “winners and losers”: some nations would gain relative to a world without climate engineering, some would lose, and none would be unaffected.<sup>85</sup> The result could be cross-border conflicts, leading to economic contraction.

**Societal risks.** Considerable amounts of SRM must take place over a longer time,<sup>86</sup> and this requires substantial finances. It is possible that non-state actors will start SRM on their own, either because they see climate change as the ultimate threat or, conversely, as a business opportunity. One company is already trying to develop a business out of SRM. A first small-scale trial of a start-up to show a “proof of concept” to later sell cooling credits has already taken place.<sup>87</sup> The cooling credits could be sold as offsets to GHG emitters.

**Legal and regulatory risk.** Solar geo-engineering proposes a regime of intentional, planetary climate modification. In theory this would require a planetary governance body, which does not currently exist. Instead, international law is decentralised, relying on state cooperation, obedience and internalisation of those rules for their enforcement.<sup>88</sup> There are some international treaties<sup>89</sup> – most notably the Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques (ENMOD),<sup>90</sup> – but none address the issue fully. Hence, it is not clear what kind of legal framework would apply in a case of insured losses due to SRM being disputed, be it between states, private entities or all these parties. Issues like this are already important to losses, as demonstrated in current cases of climate change litigation.<sup>91</sup>

**Missing out on benefits of lowering GHG concentrations in the atmosphere.** Cooling the earth via SRM may lower the earth's temperature. But SRM does nothing to reduce GHG concentrations in the atmosphere.<sup>92</sup>

With acknowledgment of the very large costs associated with climate change, governments are looking into SRM to assess technical feasibility. Academic research is ongoing in Canada, India, Japan, Australia, Germany and the UK. The US is considering a five-year study into the technologies.<sup>93</sup> Research programs in China are also modelling the climate effects.<sup>94</sup> Insurers should follow these developments carefully. Global warming can be expensive: the same could be true for man-made global cooling.

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<sup>75</sup> “The Risk of Termination Shock From Solar Geoengineering”, *Earth's Future* 6 (456–467), 2018.

<sup>76</sup> “Global Effects of Mount Pinatubo”, NASA earth observatory, 15 June 2011.

<sup>77</sup> Felgenhauer, T., et al., “Solar Radiation Modification: A Risk-Risk Analysis”, Carnegie Climate Governance Initiative, 2022

<sup>78</sup> What is the Atlantic Meridional Overturning Circulation (AMOC)? (noaa.gov)

<sup>79</sup> MacMartin, D. G., et al., “Scenarios for modelling solar radiation modification”, *PNAS* 119, 2022.

<sup>80</sup> Keys, P. W. et al., “Potential for perceived failure of stratospheric aerosol injection deployment”, *PNAS* 119, 2022.

<sup>81</sup> Carlson, C. et al., “Solar geoengineering could redistribute malaria risk in developing countries *Nature*”, *Nature Communications* 13, 2022.

<sup>82</sup> “Unregulated solar geoengineering could spark droughts and hurricanes, study warns”, *Carbon Brief*, 2017.

<sup>83</sup> Zarnetske, P., et al., “Potential ecological impacts of climate intervention by reflecting sunlight to cool”, *PNAS* 118, 2021.

<sup>84</sup> “Nature and the Insurance Industry: Taking action towards a nature-positive economy”, The Geneva Association, 2022.

<sup>85</sup> “Climate Control”, RAND Corporation, 2021, <https://www.rand.org/pubs/perspectives/PEA1133-1.html>; “Geoengineering: Reining in the weather warriors”, Chatham House, 2022.

<sup>86</sup> Baur, S. et al., “Deploying Solar Radiation Modification to limit warming under a current climate policy scenario results in a multi-century commitment”, *Earth Syst. Dynam.* 14 (367–381), 2023.

<sup>87</sup> “A startup says it's begun releasing particles in the atmosphere, in an effort to tweak the climate”, *MIT Technology Review*, 24 December 2022.

<sup>88</sup> Corbett, C., “The climate emergency and solar geoengineering”, *Harvard Environmental Law Review* 46 (97–260), 2022.

<sup>89</sup> “Response to Questionnaire on the impact of new technologies for climate protection on the enjoyment of human rights”, Heinrich Böll Foundation, Center for International Environmental Law, Action Group on Erosion, Technology, and Concentration, and Third World Network, 2022.

<sup>90</sup> “Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques (ENMOD)”, United Nations, 1978.

<sup>91</sup> Setzer, J. and Higham, C., “Global trends in climate change litigation: 2022 snapshot”, Grantham Research Institute on Climate Change and the Environment and Centre for Climate Change Economics and Policy, London School of Economics and Political Science, 2022.

<sup>92</sup> Baur, S. et al., “Deploying Solar Radiation Modification to limit warming under a current climate policy scenario results in a multi-century commitment”, *Earth Syst. Dynam.* 14 (367–381), 2023.

<sup>93</sup> “New Report Says U.S. Should Cautiously Pursue Solar Geoengineering Research to Better Understand Options for Responding to Climate Change Risks”, National Academies for Science Engineering and Medicine, 2021.

<sup>94</sup> “Climate Control”, RAND Corporation, 2021, <https://www.rand.org/pubs/perspectives/PEA1133-1.html>



# Toxic chemicals – contamination claims on the horizon

Advances in attribution science of chemicals and a change in environmental regulations have the potential to trigger property damage or bodily injury in the present day on account of chemicals having made it into the environment way back in the past.

## Potential impacts

- New scientific methods and computation power are making it easier to attribute chemicals to their sources and to discover previously not visible connections between chemicals, environmental impairment, property damage and/or bodily injury.
- In the context of strengthening legislation and regulation, these findings can lead to claims in general liability, directors and officers', and workers' compensation insurance.
- The ubiquity and accumulation potential of certain chemicals over the long term makes this a long-latent and serial-loss damage type of risk. Stakeholders across the chemical production, distribution, use and disposal value chain could be impacted.

## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



Chemicals introduced decades ago are being scrutinised for their potential to harm people and the environment in the present day. The reasons are manifold and the associated factors have evolved over many years. Scientific progress and technology have strengthened capabilities to discover if an exposure to a chemical comes from a specific source. Compensation for losses in the case of bodily injury or physical damage can be more easily secured where a source is identified.

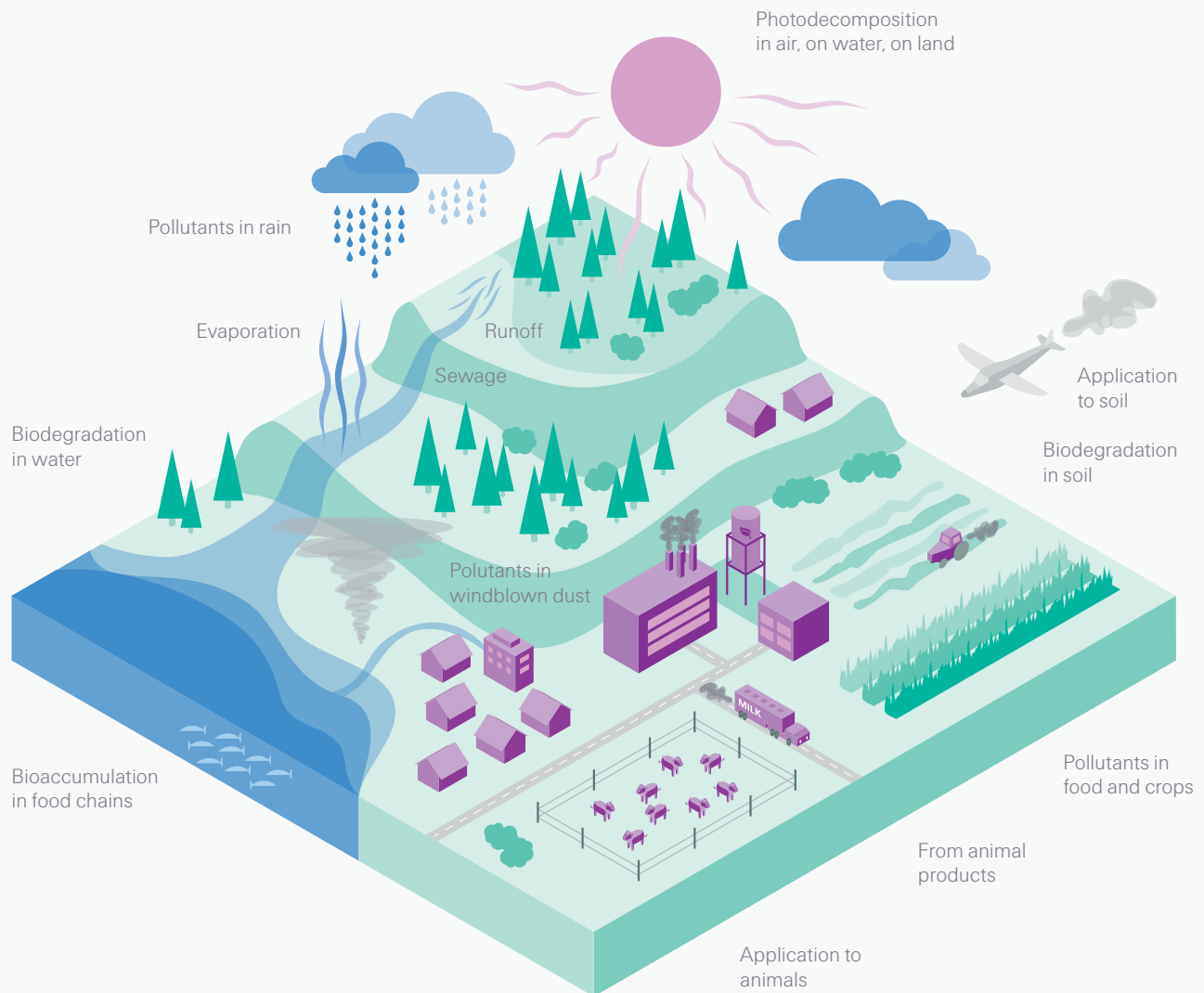
## More sophisticated tools

Today scientific tools are better able to measure the levels of chemicals in the environment, the food chain and in humans. This applies not only to single compounds but

also to chemical mixtures. In the past, most research evaluated the risk of individual substances. Due to their complexity and the lack of data, knowledge of the toxicity of chemical mixtures was limited. Today, state-of-the-art techniques enable the rapid analysis of chemical compounds in mixtures. Combining the findings with computational models facilitates improved understanding of the toxicity of chemical compounds and their impact on the environment.<sup>95</sup>

One example of this are Machine Learning (ML) techniques, which can be deployed to estimate the effects of chemicals on the environment. Recent ML achievements in predicting biological responses and environmental behaviour of chemicals are impressive. They include aspects like

## Persistent environmental contaminants



### Exposure – Accumulation – Bioconcentration

When considering contaminants in the environment, three terms are of key importance:

**Exposure:** in biological or medical terms, exposure refers to the potential contact of an organism with an agent of harm. In the context of contaminants in the environment, the following aspects of exposure are to be considered:

- Damages triggered by the physical properties of a contaminant. Small particles in air-pollution or asbestos are good examples. With physical contact, they damage lung tissue. And the matter is small enough to pass through the bloodstream and damage other organs. Radiation also belongs in this group. It damages the genome of a cell (radioactivity) or tissue directly (heat).
- The chemical properties of a contaminant that interfere with the metabolism of an organism. An example here are endocrine disruptors mimicking hormones, or mercury disrupting the metabolism of the brain.

**Accumulation:** mercury is good example to highlight the dangers of accumulation. Mercury in the environment in large part originates from human activity such as coal fired power plants, metal mining and processing, cement production or waste

disposal. As a heavy metal, it accumulates on surfaces and the in soil. Subsequent transport via wind or water distributes it into rivers, oceans and other areas. Humans can ingest mercury directly or be exposed to accumulated forms in the food chain.

**Bioconcentration:** this is when organisms ingest contaminants faster than they can excrete or metabolise them into a harmless form. That leads to an accumulation of contaminants in food sources. If the food is ingested by another organism, it also consumes the accumulated concentrations of contaminants. Over time and many steps, this can lead to high exposures of contaminants at the top of the food chain. The process is also called biomagnification.

Source:

[Definition of environmental exposure | NCI Dictionary of Cancer Terms – NCI](#)

[Biomagnification and Bioaccumulation | National Geographic Society](#)

bioactivity (as relates to both toxicity and efficacy depending on the compound class and the living system impacted) and fate/persistence (in the body and in the environment). Based on large databases, ML algorithms can generate novel, testable and verifiable assumptions, facilitating more predictable results.<sup>96</sup>

Recent advances in modelling and the use of ML also open the door to attributing chemicals in the environment to point sources.<sup>97</sup> The ML methodology has been used in processing data from satellites to obtain ground-level concentrations of atmospheric pollutants, pollution source apportionment, and spatial distribution modelling of water pollutants.<sup>98</sup>

### Transparency and standardisation

The technological developments come parallel to efforts to establish global standards for the assessment of harmful chemicals and knowledge exchange. For example, the OECD is coordinating a Hazard Assessment Program for industrial chemicals;<sup>99</sup> the Environment Programme of the United Nations (UNEP) provides a searchable chemical regulation database;<sup>100</sup> the REACH Regulation<sup>101</sup> of the European Union (EU) and the EU's Sustainability Strategy for Chemicals<sup>102</sup> contain a future risk assessment component for chemical mixtures; and the aim of the UN's Globally Harmonized System of Classification and Labelling of Chemicals<sup>103</sup> is transparency around hazardous chemicals for all.<sup>104</sup>

At the same, time legal changes to existing and the creation of new legislation can facilitate claims for property damage, environmental impairment, bodily injury and subsequent financial loss. Examples include the EU's review of the Environmental Liability Directive,<sup>105</sup> the aim of which is to protect natural resources, or the EU Directive on Representative Actions<sup>106</sup> for the protection of the collective interests of consumers. These will bring class-action type claims to the EU. In certain jurisdictions, litigation by shareholders is also a possibility. This includes claims against corporates for not adequately disclosing chemical exposures or providing timely information to investors.<sup>107,108</sup>

With these initiatives, in the future the connection between harm to people or the environment and a chemical may be easier to establish. This generates a new risk landscape. Appropriate underwriting measures in respect to policy wording (eg, claims made triggers, cost within the limit etc) are important tools to curtail future exposures. Here classical liability policies and potentially D&O covers could be in focus.

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<sup>95</sup> Luo, Y. et al., “Chemical and biological assessments of environmental mixtures: A review of current trends, advances, and future perspectives”, *Journal of Hazardous Materials*, 2022.

<sup>96</sup> Wu, X. et al., “Machine learning in the identification, prediction and exploration of environmental toxicology: Challenges and perspectives”, *Journal of Hazardous Materials*, 2022.

<sup>97</sup> “Where did the PFAS in your blood come from? These computer models offer clues”, *Environmental Health News*, 28 November 2022.

<sup>98</sup> Lu, X. et al., “Data-Driven Machine Learning in Environmental Pollution: Gains and Problems”, *Environ. Sci. Technol.*, 2022.

<sup>99</sup> “Chemical Safety and Biosafety Progress Report”, OECD, 2022.

<sup>100</sup> “Global Chemical Regulations Database”, UNEP (accessed 19th April 2023).

<sup>101</sup> “REACH”, EU (accessed 19th April 2023).

<sup>102</sup> “Chemicals Strategy for Sustainability”, European Chemicals Agency (ECHA).

<sup>103</sup> “Globally Harmonized System of Classification and Labelling of Chemicals (GHS)”, UNECE.

<sup>104</sup> “Globally Harmonized System of Classification and Labelling of Chemicals (GHS)”, UNECE.

<sup>105</sup> “Environmental Liability Directive”, EU (accessed 19th April 2023).

<sup>106</sup> “Injunctions Directive and Representative Actions Directive”, EU (accessed 19 April 2023).

<sup>107</sup> “Ex-PG&E Execs Paying US dollars 117M To Settle California Wildfires Lawsuit”, *Insurance Journal*, 2022.

<sup>108</sup> “Form 10-Q For the quarterly period ended September 30”, PG&E Corporation, 2020.





# Sustainable recycling – emerging liabilities in the circular economy

Product and associated manufacturing processes that are not circular-economy compliant present a new risk for insurers.

## Potential impacts

- Vertical or horizontal integration of producers in the circular economy may lead to new risk accumulation and supply chain dependencies.
- If recyclability becomes a mandate, any product design not fitting to a legal framework could be regarded as faulty, triggering product liability and/or recall losses.
- Companies could also be sued by workers in the recycling industry for workplace-related injuries when, in terms of design, products are not safe for recycling.
- Defence costs for such lawsuits would impact insurance policies before there is clarity about claims being valid in the first place.
- Investors could file claims against a company and its directors and officers for wrongful disclosures relating to products that are not circular-economy compliant.
- If products are no longer approved for marketing because they are not circular-economy friendly, their business viability may decrease. This could impact investment values.
- In the long run, claims for environmental impact could go down because with recycling, the disposal phase of product lifecycle is avoided.
- Property and site related casualty exposures could increase due to storage of flammable and hazardous materials in circular economy facilities (eg, storage units)

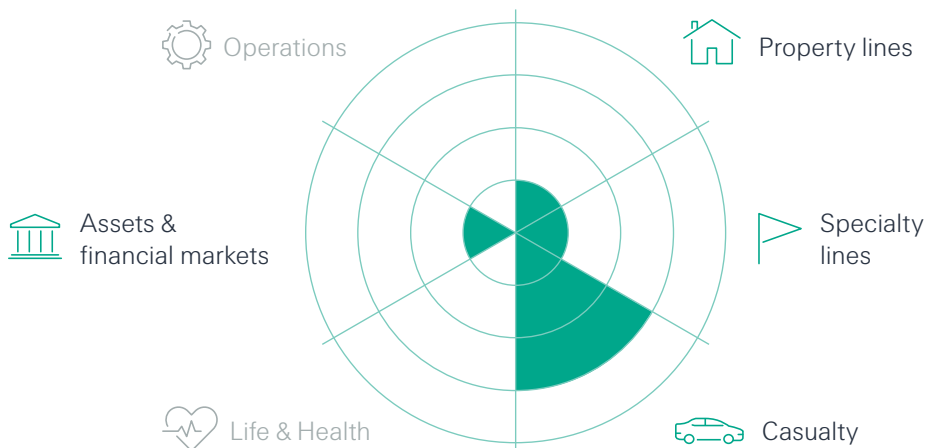
## Impact



## Time frame

0 – 3 years  
> 3 years

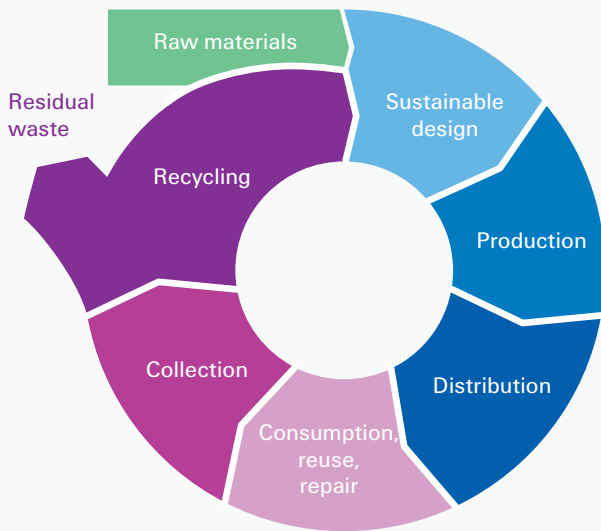
## Impacted business areas



A circular economy is a market in which products are recycled rather than scrapped, avoiding also the extraction of new resources for new products. In a circular economy, all forms of waste, be that textiles, scrap metal and obsolete electronics are returned to the economy or used more efficiently.<sup>109</sup> If what today is deemed waste is instead turned into a raw material or a new product, where does the responsibility for product safety of the original producer end, and where does responsibility for a recycled product transfer to the new producer?

The question will become more pressing as the circular economy becomes more commonplace. The European Union (EU) is drafting new legislation in the context of the Green Deal, which aims to make Europe the first climate-neutral continent.<sup>110</sup> Part of this is the *Circular Economy Action Plan* directed at sustainable growth.<sup>111</sup> Among others, the drafts include proposals for substantiating green claims made by companies,<sup>112</sup> revisions to regulations on packaging waste,<sup>113</sup> the *Ecodesign for Sustainable Products Regulation*,<sup>114</sup> and a proposal for sustainable and circular textiles.<sup>115</sup> The intention is to hold firms accountable for the sustainability claims they make, and to provide

The circular economy model:  
less raw material, less waste,  
fewer emissions



transparency for consumers and investors. This will change the liability risk landscape.

A circular economy can have implications for how existing legislation is applied. For example, a circular economy requires that products can be recycled. Already now, there are discussions about whether the *Product Liability Directive* of the EU, which requires product safety over its entire life cycle including disposal or recycling, should also apply to damages triggered by products that are not safe to recycle. The latter is often the case with products that have long life cycles. At the time of their development, it may not be foreseeable that one day a product would be recycled, and the original design may not take this future scenario into account.

Case law outside the disposal field provides examples that if producers can see the product will be dealt with in a certain way, they may be held liable.<sup>116</sup> Hence, if a recycling worker is injured during the phase of dismantling a product so that it can be recycled, the producer could be liable for injuries if the dismantling process was not considered in the original product design.

This kind of cause and effect is not restricted to workplace injuries. In the US, several states have started to regulate waste at the consumer goods level by imposing “circular economy” directives.<sup>117</sup> Hence, the city of Baltimore is suing cigarette manufacturers for costs born of litter in the city. The losses include the cost of clean-up and disposal of cigarette filters, damage to natural resources and depreciation of property values. And, based on violation of existing state and local pollution laws,<sup>118</sup> a manufacturer

can also be subject to fines for dumping by consumers of filter litter in the city.<sup>119</sup>

The circular economy is gaining traction globally. For instance, Canada has adopted a law to minimise plastic waste, Ecuador has an *Organic Law of Inclusive Circular Economy*, China a *Circular Economy Promotion Law*, and Egypt a *National action plan for sustainable consumption*, to name a few.<sup>120</sup> As momentum gathers, insureds need to make their products circular-economy ready to avoid potential claims under individual countries’ liability laws. This involves considering the entire life cycle of a product, including distribution-related responsibilities. Just because a product can be recycled does not mean it will be recycled (eg, packaging, plastic bottles). This can be an issue when the law mandates that manufacturers recycle all materials.

Whole logistics concepts will have to be rethought. For insurers, this means considering what they cover under product liability business and also how the required materials, parts and sub-components are sourced. And as insureds, manufacturers will need to deepen understanding of their own processes and supply chains to avoid cases of oversight. Careful risk assessment is becoming a new frontier in product and process design.

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- “The sharing economy”, SONAR 2015

<sup>109</sup> “Circular Economy”, UNCTAD (accessed 19 April 2023).

<sup>110</sup> “A European Green Deal”, European Commission (accessed 19 April 2023).

<sup>111</sup> “Circular economy action plan”, European Commission (accessed 19 April 2023).

<sup>112</sup> “Environmental performance of products & businesses – substantiating claims”, European Commission (accessed 19 April 2023).

<sup>113</sup> “Packaging waste”, European Commission (accessed 19 April 2023).

<sup>114</sup> “Sustainable products”, European Commission (accessed 19 April 2023).

<sup>115</sup> “Textiles strategy”, European Commission (accessed 19 April 2023).

<sup>116</sup> Ulfbeck, V. “From Linear to Circular Value Chains: A Role for Tort Liability in Recycling Practices?”, *European Journal of Risk Regulation*, 2022.

<sup>117</sup> “ESG Litigation Update: Circular Economy-Focused Litigation”, ArentFox Schiff, 2023.

<sup>118</sup> “ESG Litigation Update: Circular Economy-Focused Litigation”, ArentFox Schiff, 2023.

<sup>119</sup> “City of Baltimore Files a First of its Kind Lawsuit Against Tobacco Companies for Cigarette Filter Waste”, Mayor Brandon M. Scott, Baltimore City, 2022.

<sup>120</sup> “Circular economy: Navigating the evolving global policy landscape”, EY, 2022.



# Digital payments – disruption risks in a cashless economy

The COVID-19 pandemic propelled the use of digital payments with people embracing online shopping as never before. What risks does the cashless economy bring?

## Potential impacts

- Claims related to malfunction of software and hardware systems at point of purchase/transaction are likely to increase.
- Liability claims related to power outages, cyber attacks and natural disasters that result in digital payment system shutdowns or malfunctions may occur.
- Even a seamlessly operational digital/cashless economy entails the risk of discrimination claims. These could come from consumer protection/interest groups seeking compensation for individuals/shares of populations excluded from the market because they still use cash.
- Ability of customers to rapidly transfer money digitally increases the risk of bank runs and can have ripple effects on the financial markets.
- A shift from physical burglary to money-transfer crime.
- A cashless economy creates a new product segment for cyber insurance.

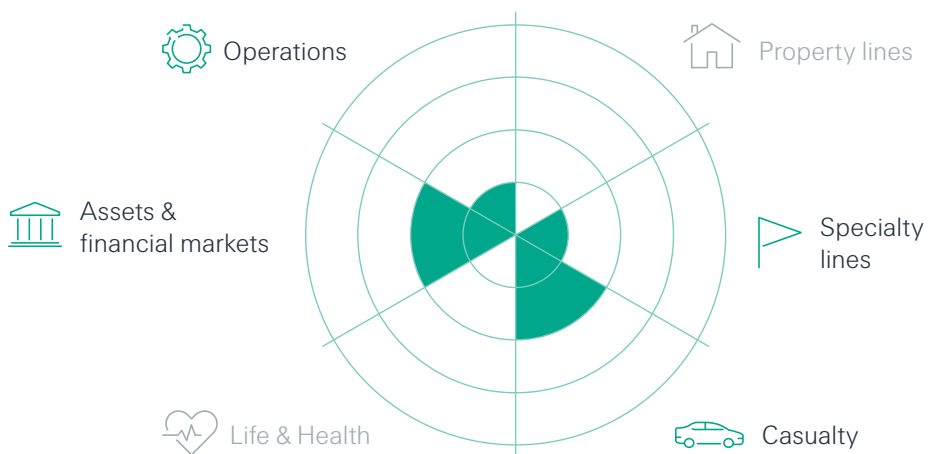
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



Physical currencies have been the mode of economic transaction for thousands of years. Today, payment systems are becoming increasingly cashless. In the European Union (EU), one in five of all daily purchases in 2022 took place online, and one-third were made using a card.<sup>121</sup>

The preference to use cash or digital methods varies. For example, in Sweden 15% of all in-person purchases were made in cash 2019; in Greece the figure was 88%.<sup>122</sup> All told, the trend for cashless payments is set to accelerate as the digital revolution goes increasingly global. For instance, in Sweden already only half of bank branches accept cash deposits and in 2022, Europe's leading six largest online-only banks had more than 34 million customers.<sup>123</sup> In parallel, emerging markets are digitalising their economies,

transitioning from cash to digital payments and bypassing the intermediate "card age".<sup>124</sup> In Kenya, more people use mobile money transfer applications than own a bank account.<sup>125</sup> In many nations in Sub-Saharan Africa, mobile money accounts are more common than bank accounts.<sup>126</sup>

## Systemic benefits...

De-cashing the economy could bring many benefits. With the elimination of notes and coins, a larger share of money supply would be in form of virtual bank deposits. If banks have access to a larger digital money supply, they can increase the volume of loans they make and improve lending costs. In addition, going cashless would make the economy more responsive to central bank monetary policy

changes, as cash at home is less sensitive to changes in interest rates than bank deposits. It should also improve fiscal revenues as tax-evasion becomes harder.

### ...and new risk pools

However, digital payments require infrastructure: for example a network, electricity and functioning hardware (phone, smartphone, card etc). Failure in any of these can paralyse a whole economy. At the consumer level, payment system outages would mean transactions at a restaurant or the cashless check-out in a supermarket, among others, would be blocked. Further, it could be that salaries are not/cannot be paid, or that an entire payments system is down for days. Major disruptions triggered by cyberattacks, power outages and natural disasters are a main threat to the cashless economy. But they also bring new risk pool opportunities like cyber and digital fraud insurance.

Another risk is that many cashless/digital payment systems are privately run. This leaves households dependent on private players to access money and for day-to-day transactions. In case of a private payment company going bankrupt, the guarantee of access to one's e-wallets may be at risk. With individuals losing the ability to save in physical currency, their virtual savings might be more vulnerable to financial crises or collapse of private banks. And some sectors that rely heavily on cash transactions (eg, housing construction) may struggle during transition to cashless.<sup>127</sup> This could see a rise of new a business for re/insurers: coverage for deposit guarantees designed for non-bank payments providers, and digital wallet insurance.

A cashless economy will exclude some parts of the population that are not digitally equipped. Everyday needs like purchasing a bus ticket will be difficult to access without a card or smartphone. This could lead to consumer

protection groups filing discrimination lawsuits against private companies, municipalities and/or the state. A cashless economy also brings with it a certain loss of personal privacy. Digital payments leave digital traces, as personal data is shared and stored across multiple vendors and platforms. There are examples of where digital credit decisions are taken based on unclear data and non-transparent processes.<sup>128</sup> Cyber security incidents resulting in leaked personal data are more common when more personal data is stored by financial institutions.<sup>129</sup> Further, financial crimes based on social engineering are reportedly increasing with growing online access to personal data.<sup>130</sup>

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<sup>121</sup> “Cash remains the most frequently used means of payment in stores but electronic payments grow further, ECB study shows”, European Central Bank (ECB), 2022

<sup>122</sup> “Access To Cash Review Final Report”, Access to cash review, 2019, p 55..

<sup>123</sup> Erlanger, S., “A Cashless Economy: How to Protect the Low-Income”, Cardozo Law Review de•novo, 2019, p 180; “Number of customers at largest online only European banks as of 2022”, Statista, 2023.

<sup>124</sup> “Deposit Insurance Treatment of E-Money”, Consultative Group to Assist the Poor (CGAP), 2019, p 1; “Navigating the payments matrix” PwC, 2021, p 6.

<sup>125</sup> Erlanger, S., “A Cashless Economy: How to Protect the Low-Income”, Cardozo Law Review de•novo, 2019, p 178.

<sup>126</sup> “Deposit Insurance Treatment of E-Money”, Consultative Group to Assist the Poor (CGAP), 2019, p 1.

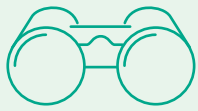
<sup>127</sup> “The Macroeconomics of De-Cashing”, IMF Working Papers, 2017.

<sup>128</sup> “Credit Scoring Approaches Guidelines”, The World Bank Group, 2019, p 26 (accessed 28 March 2023).

<sup>129</sup> “Personal Data use in Financial Services and the Role of Financial Education”, OECD, 2020, p 18 (accessed 28 March 2023).

<sup>130</sup> “The Risks of Embodying ‘Cashless Society’ in Indonesia”, Center for Digital Society, 2020, p 6–7.





## Digital twins: virtual product testing and maintenance

Digital twins are being integrated across all industries, yielding cost-savings in product testing, design and maintenance. However, widespread use of this technology, especially in combination with artificial intelligence, could increase the risk of systemic product liability and recall claims. It also raises concerns about privacy and cybersecurity.

### Digital twins are becoming a reality in most sectors...

By bridging the gap between the virtual and the real worlds, digital twins are powering a “fourth industrial revolution”. The twins are digital replicas of physical objects (eg, a car) or systems (eg, a production line). They enable virtual modelling to analyse and optimise real-world products and processes.

Use of digital twins is becoming common in all sectors. In urban planning, they are used to model traffic flows, energy use and carbon emissions. In the automobile industry, they allow researchers to run crash-test simulations countless times without having to destroy cars. They also support sustainability initiatives: for example, using “Virtual Singapore”, a digital twin of the city state, researchers determine where best to build solar panels for power generation by analysing light and temperature variations across different locations.<sup>131</sup>

Digital twins can facilitate cost-reduction and operational efficiencies. They can be used to plan maintenance, spot emerging problems, and simulate the effect of upgrades and design changes. By allowing researchers to virtually test multiple versions of new models before producing a physical prototype, the twins can help reduce the incidence of product defects and also time to market.<sup>132</sup> In manufacturing, they help with training of personnel and production facility planning.<sup>133</sup> And, once a product is in use, predictive maintenance helps reduce downtime: the digital twin can be fed with real-time data to monitor and simulate product behaviour in different environments. This can help expand the scope within which the product can operate and also prolong its lifetime.<sup>134</sup>

### ... and are expected to reduce claims across a range of business lines

The versatility of digital twins and their rapid integration across sectors is likely to bring benefits to Property & Casualty insurance. In manufacturing, the predictive maintenance capacity of the twins is likely to reduce overall claims in product liability and recall. Data-driven, real-time tracking of key process parameters can also help reduce product quality issues. In production facility planning (or other environments with man-machine interaction), digital twins can identify potential threats to worker, contractor and visitor safety, and even long-term ergonomics, ultimately reducing employer’s liability and worker compensation claims. In motor insurance, when applied to autonomous car features, twins can help reduce accident frequency and severity. Their application to process management or supply chain optimisation can reduce business interruption risks. Digital twins of assets can also be stress-tested against a pre-selected series of events (eg, storms, earthquakes or infectious disease outbreaks), to provide risk insights and increase resilience.<sup>135</sup>

### However, widespread application could lead to systematic risk scenarios...

Given the need for supercomputing capabilities and know-how, there is a risk that the development of digital twin software becomes concentrated among a few major players. If this were to lead to deployment of standardised software across many sectors, the risk of systemic losses in the event of software fault would rise, triggering multiple product liability claims and recalls from



many manufacturers. Professional indemnity and product liability losses could be invoked for digital twin software and data providers in cases where the software or virtual assumptions are the root-cause of a fault event.

In cases where digital twin technology is integrated with the Internet of Things (IoT) and artificial intelligence (AI), there is risk of systemic losses if AI is used in real-time to provide decisions on operation of the IoT assets. This risk is high since, due to their complexity, AI decision-making processes are often untraceable. Apart from losses due to asset failure, directors and officers claims could also be triggered for losses incurred on account of management making wrong decisions informed by the outcomes of digital-twin simulation exercises.

### ... and brings many unknowns regarding data privacy and access

Creating digital replicas also raises questions around privacy and cybersecurity. The operation of many digital twins is based on information provided by a multitude of sensors that track real-world data and movements. While they enable collaboration across processes and systems, the sensors also access sensitive information.<sup>136</sup> This leaves businesses vulnerable to cyberattacks, data theft and sabotage. There are also challenges in defining data ownership and intellectual property rights, especially in the case of third-party providers of digital twin software. This is particularly relevant for insurers that rely on access to data from virtual product testing to price risk.

<sup>131</sup> "Exclusive: Inside Singapore's strategy for battling climate change", GovInsider, 2020.

<sup>132</sup> "Digital twins: adding intelligence to the real world", Capgemini Research Institute, 2022.

<sup>133</sup> "How digital twins are transforming manufacturing, medicine, and more", Time, Dec 2021.

<sup>134</sup> "Industry 4.0 and the digital twin", Deloitte University Press, 2017.

<sup>135</sup> "A world in turmoil demands digital transformation and cooperation", Swiss Re, 14 Feb 2023.

<sup>136</sup> "Digital twins: Bridging the physical and digital", Deloitte Insights, 15 January 2020.





# Generative AI

The rapid development of generative artificial intelligence (AI) systems opens opportunities for underwriting and claims processing. It also raises questions of ownership and associated risks.

Large data sets form the core of generative AI models, the latter a set of algorithms that can create seemingly realistic content like text, images, or audio using training data.<sup>137</sup>

## Who owns what?

The collection of the large data sets raises questions around ownership of the data, and also concerns about data quality and biases. Ownership is still a relatively new field but already, in the US and UK, a few lawsuits have been filed to challenge the data used in and produced by generative AI models.<sup>138</sup> Recently, a class action case was filed by several artists whose images had been used to train generative AI tools.<sup>139</sup> In the meantime, several platforms have banned AI-generated art.<sup>140</sup>

In the US and in Europe, class actions against the technology sector have been on the rise.<sup>141</sup> This poses a risk for the insurance industry as class actions can be expensive. While legal disputes, together with existing and new legislation, will help establish some guiding principles, the AI industry will likely adapt to new tech-generated disruption in similar fashion to other sectors that have been disrupted by innovation.<sup>142</sup> For example 15 years ago, the music and movie industries suffered losses and struggles over rights related to illegal downloads. Then with innovation came platforms allowing consumers to listen and view online content legally, while at the same time compensating rights holders.

## Concerns around data quality and biases

Another risk relevant for the insurance industry is that publicly-available generative AI systems can discourage people to seek professional medical, legal or financial advice. These systems make predictions using training data only. They do not access real-time data and thus may generate misleading information. Based on this information, an individual may decide not to seek, for instance, medical help when really a visit to the doctor should be the course of action. The outcome can be a worsening of health status, requiring more intense (and expensive) treatment in the future and, in turn, higher claims on health insurance policies.

The question comes down to the quality and quantity of data used in the development of generative AI systems. An example from Finland, prior the recent developments of

generative AI systems, illustrates how regulators may find the use of non-individual data discriminative. The National Non-Discrimination and Equality Tribunal prohibited a credit institution from using a decision-making method based on criteria such as gender, first language, age and residential area, the said criteria itself based on assumptions derived from general statistical data and information on payment defaults. The Tribunal stated that the creditworthiness of an applicant became weaker than it would have been using other information.<sup>143</sup> Further, several Data Protection authorities investigate complaints related to the use of and invented (non-accurate) personal data derived from generative AI systems.<sup>144</sup> In the EU, the European Data Protection Board has launched a task force to help its members on data protection issues linked to generative AI.<sup>145</sup>

If the training data is too limited, biased or is incorrect, the predictions/answers generated will likewise be biased and inappropriate. This could lead to medical malpractice claims or professional liability lawsuits. On the other hand, generative AI models can be trained on a company's proprietary knowledge base and processes. For example, insurers could deploy generative AI models to improve the efficiency of in-house manual underwriting and claims processes.

<sup>137</sup> See "Generative AI", BCG (accessed 19 April 2023).

<sup>138</sup> "Getty Images Statement", Getty Images, 2023; "AI Art Generator Copyright Litigation", Joseph Saveri Law Firm (accessed respectively 16 February 2023). "AI Drake just set an impossible legal trap for Google", The Verge, 19 April 2023.

<sup>139</sup> "GitHub Copilot litigation", Joseph Saveri Law Firm & Matthew Butterick, 3 November 2022, "Stable Diffusion litigation", Joseph Saveri Law Firm & Matthew Butterick, 13 January 2023 (accessed respectively 16 February 2023).

<sup>140</sup> "Getty Images Content License Agreement", Getty Images, August 2022; "Newgrounds Wiki – Art Guidelines", Newgrounds; "Artmageddon: The rise of the machines, and banning machine-generated images", Purpleport, 14 September 2022 (accessed respectively 16 February 2023).

<sup>141</sup> "European Class Actions Report", CMS, 2021; "Class & Group Actions", Global Legal Group, 2021.

<sup>142</sup> "After inking its OpenAI deal, Shutterstock rolls out a generative AI toolkit to create images based on text prompts | TechCrunch", TechCrunch, 25 January 2023.

<sup>143</sup> "Plenary session (voting)", National Non-Discrimination and Equality Tribunal of Finland, 21 March 2018.

<sup>144</sup> "Factbox: Governments race to regulate AI tools", Reuters 4 May 2023., "Federal privacy watchdog probing OpenAI, ChatGPT following complaint", CBC 4 April 2023.

<sup>145</sup> "EDPB resolves dispute on transfers by Meta and creates task force on Chat GPT", EDPB, 13 April 2023., "European privacy watchdog creates ChatGPT task force", Reuters, 14 April 2023.



# AI Image Generator

Request:

Generate image of a girl with a pearl earring similar to famous painting

Result:



Regenerate

Save As...



# Weight-loss injections: miracle drugs or new risks in a bottle?

Weight-loss injections are rising in popularity and show potential in the management of obesity. But uncertainty remains over long-term risks to health, and there are also concerns regarding the use of the injections for aesthetic weight loss in unintended (ie, not obese, not overweight) populations.

## Potential impacts

- Potential product liability claims against drug manufacturers, distributors and retailers, where use of the drugs lead to unexpected adverse health impacts.
- The outcome of potential class actions could go beyond the pharma industry, for instance to pharmacies and distributors also.
- Medical malpractice or professional indemnity claims against doctors and pharmacies due to use of weight-loss drugs in populations that do not meet the indication criteria.
- Potential unknown long-term health impacts could affect longevity and morbidity.
- Cosmetic use might cannibalise medical-indicated use for diabetes, causing inequity and drug shortages.

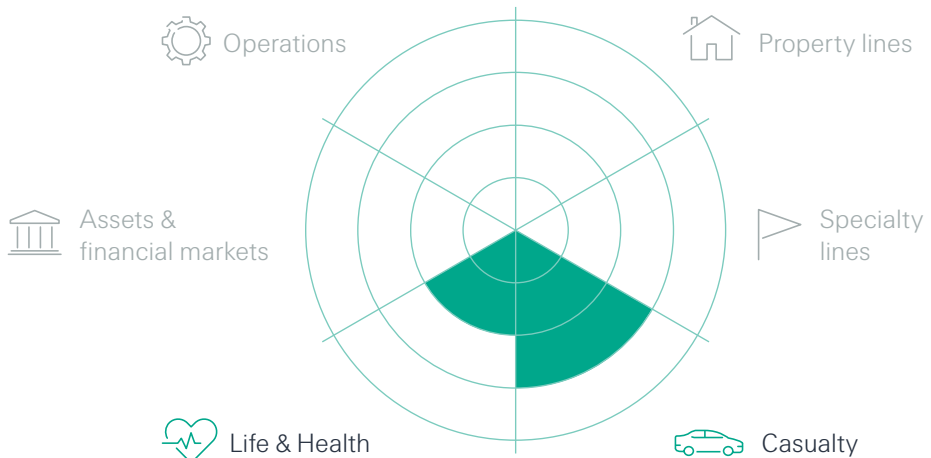
## Impact



## Time frame

0 – 3 years  
> 3 years

## Impacted business areas



A new generation of semaglutide and other GLP-1 incretin mimicking compound weight-loss drugs are gaining popularity. Influencers and celebrities publicise the merits of these drugs for rapid weight loss. Some estimates see global sales of said medications at USD 150 billion by 2031, not far off the size of today's cancer treatment market.<sup>146</sup> Beyond the use for aesthetics by those who can afford it, semaglutide injections are being hyped as a miracle cure for the obesity pandemic. "Hype", however, may well prove the defining term: the drugs also raise concerns regarding uncertain health risks.

## Why the hype?

Originally developed to treat diabetes, the drugs have been found to also reduce body weight and in the US, have recently been Federal Drug Administration-approved for such usage in overweight individuals.<sup>147</sup> In clinical trials, semaglutide injections led to average weight loss of about 15%.<sup>148</sup> The medication mimics the action of human incretin glucagon-like peptide 1 (GLP-1), a hormone produced naturally by the lining of the stomach after ingestion of food. The drugs increase insulin secretion and improve blood sugar disposal, and hence glycaemic control. They also slow the emptying of the stomach, creating a feeling of fullness that reduces appetite and food intake, in turn resulting in weight loss.

### Benefits are not without side effects

For overweight and obese individuals, the weight loss benefits of semaglutide injections might outweigh the health risks, some known, some uncertain. The drugs come with side effects: nausea, vomiting, diarrhoea, abdominal pain and constipation are among those that have been reported. There are also concerns regarding the potentially serious ramifications of the artificial creation of the feeling of satiation. The long-term effects of the consistent presence of a molecule that signals satiation, which, when produced naturally by the stomach, is present for just a few minutes after food ingestion, remain unknown. Possible suspected negative effects are (mal)functioning of the thyroid gland (eg, thyroid cancer) and the pancreas (pancreatitis).<sup>149</sup>

Also, the weight-loss benefits are not always permanent. In clinical trials, many patients who stopped taking the drug regained a large proportion of their lost weight soon after. Without complementary lifestyle and nutrition changes such as reducing carbohydrate intake and cutting calories, patients can mistakenly believe they have been cured of obesity, rather than it just being in remission. The efficacy of the drug is thus reduced, which can become a recurring issue, and more so in jurisdictions like the UK where reimbursement schemes are restricted to 2 years.<sup>150</sup>

### Concerns over non-medical use

Another important concern is the growing non-medical use of semaglutide injections. With celebrities endorsing the drug for rapid weight loss, there has been a rise in cases where the injections are prescribed as a lifestyle drug, even to patients with normal BMIs.<sup>151</sup> As part of the hype, in these cases the health risks may be downplayed, which could potentially trigger claims in medical malpractice and other professional negligence covers. Given the popularity of semaglutide drugs, there is also a risk of black markets developing, with illegal sales of secondary and counterfeit products and potential for overdoses.

The novelty, combined with uncertainty over the long-term effects of semaglutide drugs and potential for misuse, opens up the possibility for liability claims and class actions, should the drugs be found to have serious adverse health effects. Beyond the drug manufacturers, many agents in the value chain, from distributors to retailers and pharmacies, could be implicated.

All medical interventions entail benefits and risks. Semaglutide injections may bring health benefits to obese individuals, giving them the opportunity for more mobile and healthier lives. However, in mildly overweight individuals or those who use the drugs for aesthetic purposes, uncertainty around the long-term risks might outweigh the benefits. In addition, their price tag (around USD 1 000 per month in the US<sup>152</sup>) raises questions as to their cost-effectiveness in managing weight loss over time. To date, lifestyle changes and nutrition strategies that include fasting, reduced carbohydrate and calorie intake remain the safest and most cost-effective interventions for sustained weight loss and reduction of long-term mortality and morbidity risks.

### Related articles in SONAR

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- “Changeing Lifestyle”, SONAR 2013

<sup>146</sup> “New drugs could spell an end to the world’s obesity epidemic”, *The Economist*, 2 March 2023.

<sup>147</sup> “FDA Approves New Drug Treatment for Chronic Weight Management, First Since 2014”, FDA, 4 June 2021.

<sup>148</sup> Bergmann et al., “Semaglutide for the treatment of overweight and obesity: A review.” *Diabetes, Obesity and Metabolism*, January 2023.

<sup>149</sup> “Summary of risk management plan for Ozempic”, EMA, 23 February 2022;

“Summary of risk management plan for Wegovy”, EMA, 10 March 2022.

<sup>150</sup> “NICE recommends new drug for people living with obesity”, NICE, 8 February 2022.

<sup>151</sup> “What Is Ozempic and Why Is It Getting So Much Attention?”, *The New York Times*, 8 March 2023.

<sup>152</sup> “Find out the cost for Ozempic®”, NovoCare.











# Competitive and business environment



## Going digital: embedded insurance

Today's digitalisation of almost any purchase and customer interaction creates new opportunities for embedded insurance solutions. But it also comes with risks.

### Jump the classical sales channel and think new

Embedded insurance solutions have been a growing trend for years and many insurers already have said offerings in their portfolios. The concept is the real-time bundling in and sale of insurance covers as part of another product or service, bringing the risk protection directly to the consumer at the point of sale.

With digitalisation, partnerships between insurers, tech companies and non-insurers have enabled seamless addition of risk protection solutions to products that consumers access/purchase online. Some covers leverage usage-based and behavioural data to individualise risk pricing. One type of solution is when an insurer partners with a product or service company in a different sector, which wants to offer white-label and customised insurance services to its clients. The insurance is offered at time of need and by a brand that a customer already knows and trusts.<sup>153</sup> The solutions are easy to understand and customers do not have to seek external insurance advice.

### Buy a car and get the car manufacturer's insurance app

Examples of embedded insurance are policies offered or included when renting properties, vehicles or buying secondhand goods.<sup>154</sup> Another example is a car insurance available in selected US states offered through an app developed by the car manufacturer. The insurance is based on factors such as driving behaviour and close-to-real-time data that the car's device can track.<sup>155</sup> Pricing does not consider age, gender or claim history.

A Chinese "super app" is often cited as success story in embedded insurance. The app partners with around 90 insurers and offers more than 2 000 risk protection products embedded in other services.<sup>156</sup> The app company knows its customers, has access to data on purchasing behaviours and, in partnership with insurers, develops tailored insurance solutions, from L&H

coverage to product warranties. The group has become the largest online insurer in China. It has more than 500 million customers.<sup>157</sup>

### Challenges to consider...

Often embedded insurance solutions cover a low sum insured, meaning that a critical mass of insureds is required for the business to remain sustainable. Without, the low premium volumes will not offset fixed costs from insurers and their partners.

Due to the distribution model (ie, sale as part a company's core service or product), the embedded insurance offering has to be simple. Oversimplified policy terms could make both consumers and insurers more vulnerable to uncertainties or misunderstandings which could lead to more claims and litigation costs. As consumers get used to a faster and simpler process of buying insurance, the expectation builds that claims and pay-out processes follow the same speed and degree of automation. This expectation may not always be met by insurance companies.

An industry challenge is that if consumers become used to having insurance embedded in other products or services, they may not seek other forms of cover that they need, assuming they are covered when they are not. On the other hand, embedded solutions could also lead to consumer complaints on the grounds of being led into buying unnecessary insurance, or ending up being over-protected by purchasing a cover that is written into another already-owned policy, such as home insurance.

<sup>153</sup> "Embedded Insurance Gaining Steam as Insurers Build Distribution Partnerships", Best's Review, January 2022.

<sup>154</sup> "Home & Motor Insurance", iptiQ; www.omocom.insurance (accessed respectively 4 April 2023).

<sup>155</sup> "Tesla Insurance", Tesla (accessed 4 April 2023).

<sup>156</sup> "Embedded Insurance Peer Group Report", Embedded Finance & Super App Strategies, June 2022, p 15;

<sup>157</sup> "Embedded Insurance a US dollars 3 Trillion market opportunity, that could also help close the protection gap", Finnovista, 2021, p 6.







# Appendix: terms and definitions

## What is SONAR?

SONAR stands for Systematic Observation of Notions Associated with Risk. It is Swiss Re's process for identifying, assessing and managing emerging risks. Experts across the company use a web-based platform to collect early signals of emerging risks. All signals are assessed and prioritised by an emerging risk management team, which works closely with topic experts from Swiss Re's different business areas. The team serves as a catalyst for risk identification and assessment to define and implement recommendations in collaboration with the business. The findings are regularly shared internally and summarised for external audiences here.

## What are emerging risks?

We define emerging risks as newly developing or changing risks that are difficult to quantify and could have a major impact on society and industry.

## What are emerging risk themes?

Emerging risk themes illustrate potential new or changing risk developments for the insurance industry. They are mainly derived from SONAR but also draw on other sources. All themes have been assessed and edited by Swiss Re's emerging risk management experts. This report only features new emerging risk themes (ie, topics covered in previous editions are not listed again). You can retrieve prior reports from our webpage: [www.swissre.com/sonar](http://www.swissre.com/sonar).

## What is meant by overall impact?

The overall impact of an emerging risk is an indicator of the potential financial, reputational and/or regulatory effect on the insurance industry. It is assessed on a scale from high to low:

- **HIGH**  
Potentially high financial, reputational and/or regulatory impact or significant stakeholder concern
- **MEDIUM**  
Potentially medium financial, reputational and/or regulatory impact or moderate stakeholder concern
- **LOW**  
Potentially low financial, reputational and/or regulatory impact, or low stakeholder concern

## What is meant by time frame?

We divide themes into those likely to occur in less than three years and those likely to occur over a longer horizon. This assessment should not be used as an indicator of when action is needed, as some themes likely to occur in the more distant future may, nonetheless, require immediate action to prepare.

## What is meant by impact per business area?

Spider graphs indicate the potential impact on major insurance industry business areas on a scale from 0 (= no impact) to 4 (= significant impact).

## What are trend spotlights?

Text boxes throughout the report provide selective focus articles on emerging trends which could become relevant for the insurance industry and its clients. The selection of topics is non-exhaustive, and descriptions are intended as food for thought and discussion starters rather than comprehensive reviews.

## What are macro trends?

Swiss Re has identified a set of macro trends assumed to have a high impact on the insurance industry within the next five to 10 years. The macro trends featured in this report have been selected independently through expert discussions and surveys. They provide context to the emerging risk insights from the SONAR process.

### Abbreviations

|       |                                                        |
|-------|--------------------------------------------------------|
| AI    | Artificial Intelligence                                |
| AMAP  | Arctic Monitoring and Assessment Programme             |
| API   | Application Programming Interface                      |
| BI    | Business interruption                                  |
| BMI   | Body Mass Index                                        |
| C&S   | Credit and Surety                                      |
| CCT   | Cirrus Cloud Thinning                                  |
| D&O   | Directors and Officers                                 |
| DSA   | Digital Services Act                                   |
| ESG   | Environmental, Social and Governance                   |
| EU    | European Union                                         |
| FDA   | Food and Drug Administration                           |
| GBAM  | Ground-based Albedo Modification                       |
| GDP   | Gross Domestic Product                                 |
| GHG   | Greenhouse Gas                                         |
| IoT   | Internet of Things                                     |
| IP    | Intellectual Property                                  |
| L&H   | Life and Health                                        |
| ML    | Machine Learning                                       |
| NATO  | North Atlantic Treaty Organization                     |
| OECD  | Organisation for Economic Co-operation and Development |
| P&C   | Property and Casualty                                  |
| PFAS  | Per- and Polyfluorinated Alkyl Substances              |
| PV    | Political Violence                                     |
| SONAR | Systematic Observation of Notions Associated with Risk |
| SRM   | Solar Radiation Management                             |
| UK    | United Kingdom                                         |
| UN    | United Nations                                         |
| UNEP  | United Nations Environment Programme                   |
| US    | United States (USA: United States of America)          |
| USD   | US Dollar                                              |
| WHO   | World Health Organisation                              |

**Title**

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